



WIPRO CAPSTONE PROJECT REPORT

Name: Nitish Kumar Sharma

University Roll no: 2444440050026

College Roll no: 33

College Id: 444-9505

BATCH: BSCIT-B1-3RD

Unit 1: VM Installation & Configuration

1.1 Agenda

- I. Choose a Virtualisation Platform
 - II. Create a New Virtual Machine Install Windows 10 or 11
 - III. Apply Updates
 - IV. Standardise the Environment
 - V. User Management:
 - VI. Security with Registry
-

1.2 Introduction

Virtualization is a technology that allows multiple operating systems to run on a single physical computer by creating virtual machines. In this unit, VMware Workstation was used to build a virtual platform and install Windows 11 Enterprise.

1.3 Software and Hardware Requirements

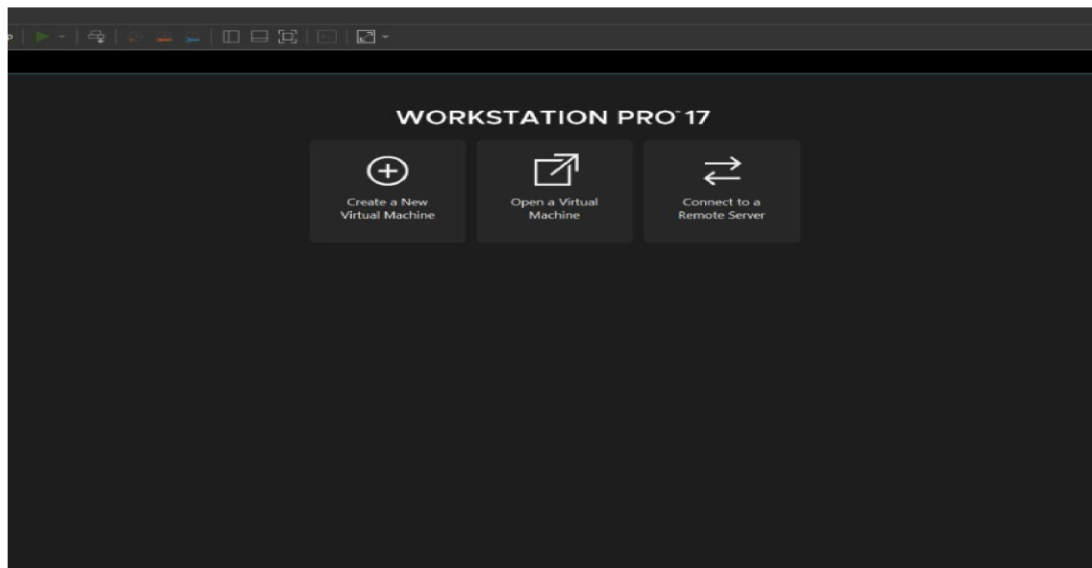
Software

- VMware Workstation
- Windows 11 Enterprise (90-Day Evaluation Version)
- Host Operating System: Windows

Hardware

- Minimum 8 GB RAM
- 64-bit Processor
- 50 GB Free Disk Space

Image 1: VMware Workstation Main Screen

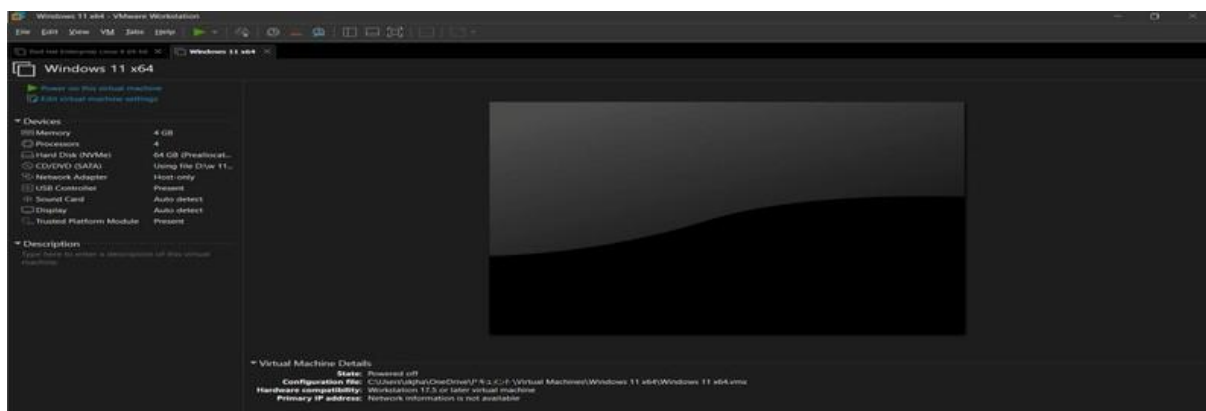


1.4 Steps

Step 1: Create Virtual Machine

- Open VMware Workstation
- Click **Create New Virtual Machine**
- Select the **Windows 11 Enterprise ISO file**
- Allocate CPU, RAM, and storage
- Start the virtual machine

Image 2: VMware Virtual Machine Wizard



Step 2: Install Windows 11 Enterprise

- Boot the virtual machine using the ISO file
- Select language and regional settings
- Choose installation disk
- Complete the Windows installation process

Image 3: Boot Manager Screen

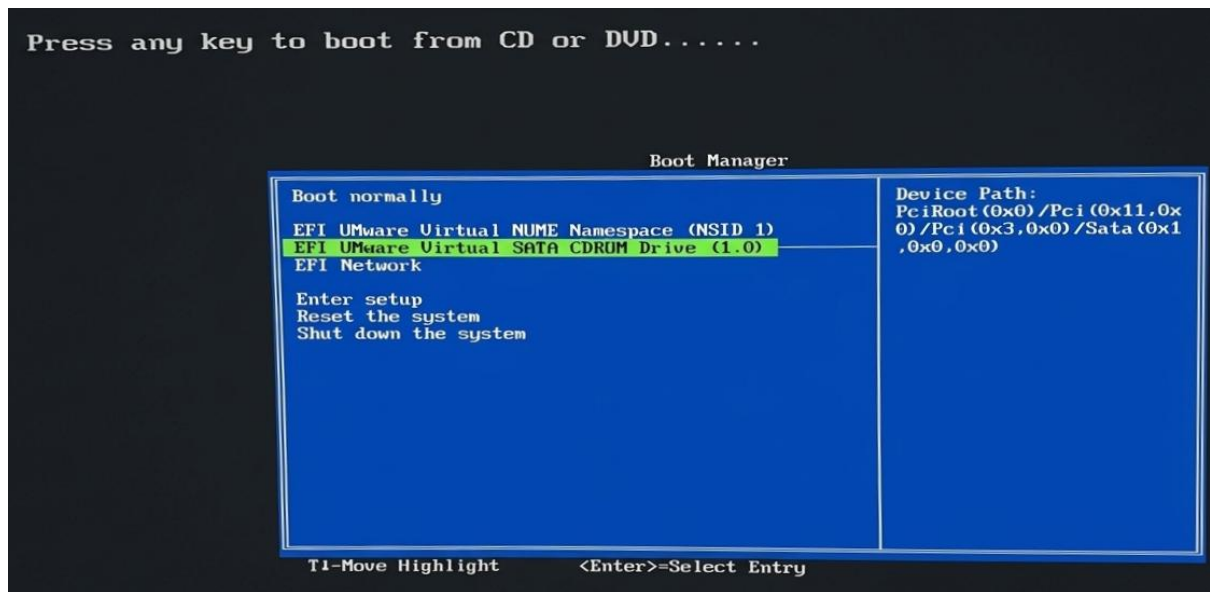
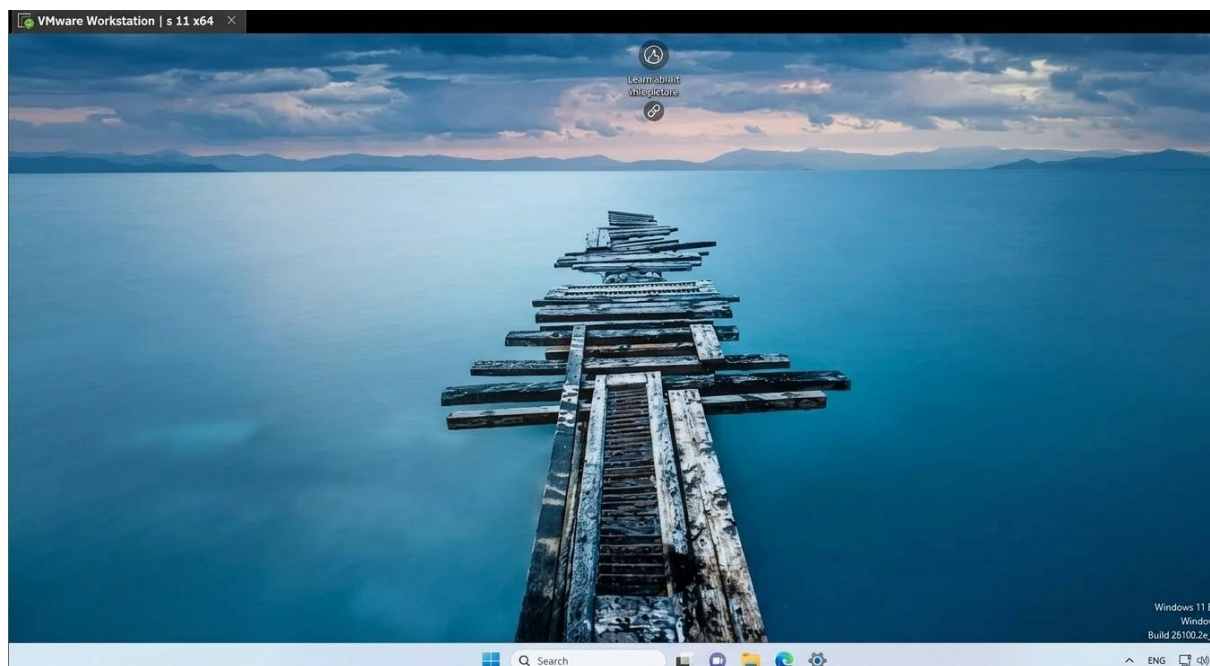


Image 4: Windows Desktop After Installation



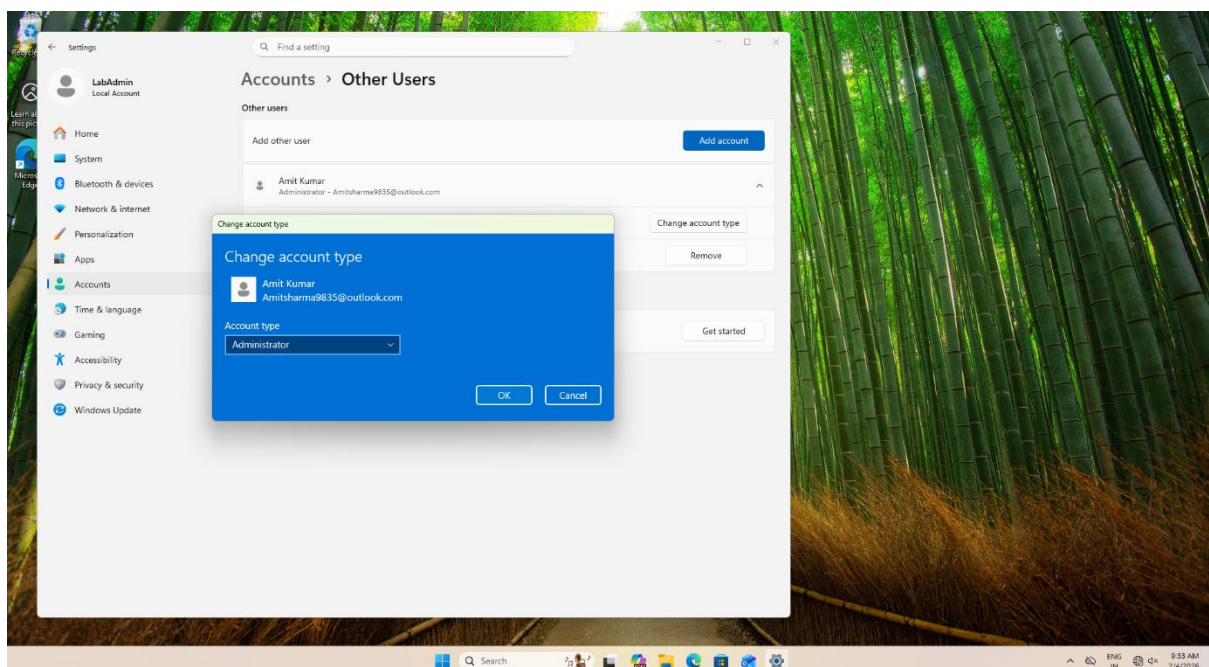
Step 3: Create User Accounts

- Open **Settings** → **Accounts** → **Other Users**
- Add a new user account
- Change the account type to **Administrator**

Two accounts were created:

- Administrator Account
- Student User Account

Image 5: Other Users Screen



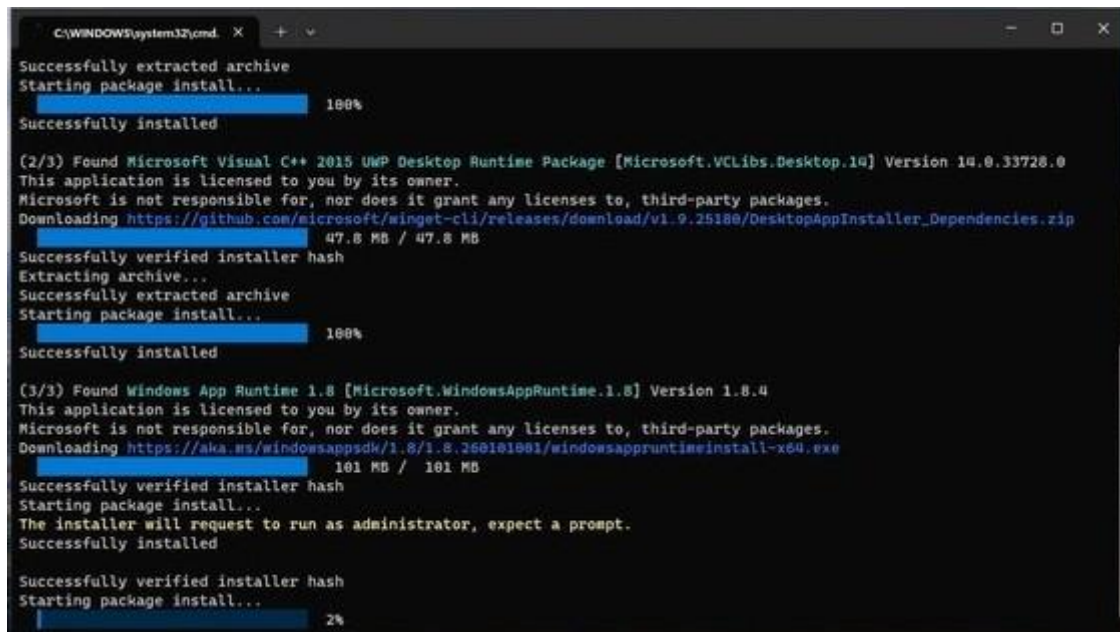
Step 4: Update System Using Winget

- Open **Command Prompt as Administrator**
- Run the following command:

“winget upgrade –all”

This command updates installed applications and system components.

Image 6: Command Prompt Running Winget



```
C:\WINDOWS\system32\cmd. X + -
Successfully extracted archive
Starting package install...
100%
Successfully installed

(2/3) Found Microsoft Visual C++ 2015 UWP Desktop Runtime Package [Microsoft.VCLibs.Desktop.14] Version 14.0.33728.0
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/microsoft/winget-cli/releases/download/v1.9.25180/DesktopAppInstaller_Dependencies.zip
47.8 MB / 47.8 MB
Successfully verified installer hash
Extracting archive...
Successfully extracted archive
Starting package install...
100%
Successfully installed

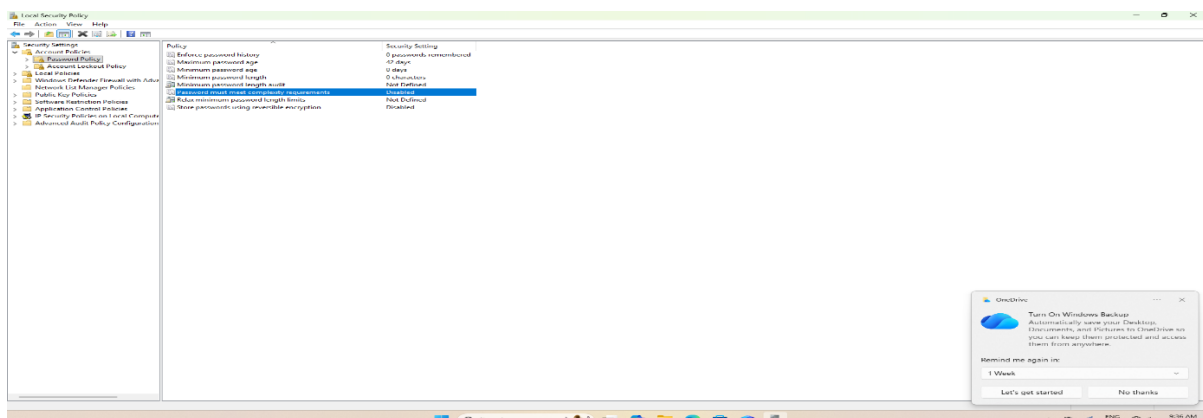
(3/3) Found Windows App Runtime 1.8 [Microsoft.WindowsAppRuntime.1.8] Version 1.8.4
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://aka.ms/windowsappsdk/1.8/1.8.368101001/windowsappruntimeinstall-x64.exe
101 MB / 101 MB
Successfully verified installer hash
Starting package install...
The installer will request to run as administrator, expect a prompt.
Successfully installed

Successfully verified installer hash
Starting package install...
2%
```

Step 5: Configure Password Security Policy

- Open **Local Security Policy**
- Navigate to **Password Policy**
- Configure the following settings:
 - Minimum password length: 8 characters
 - Password complexity: Enabled
 - Maximum password age: 42 days

Image 7: Password Policy Settings



Step 6: Restrict USB Storage Access

- Open **Group Policy Editor (gpedit.msc)**
- Navigate to **Computer Configuration → Administrative Templates → System → Removable Storage Access**
- Enable **Deny All Access**

Image 8: USB Restriction Policy

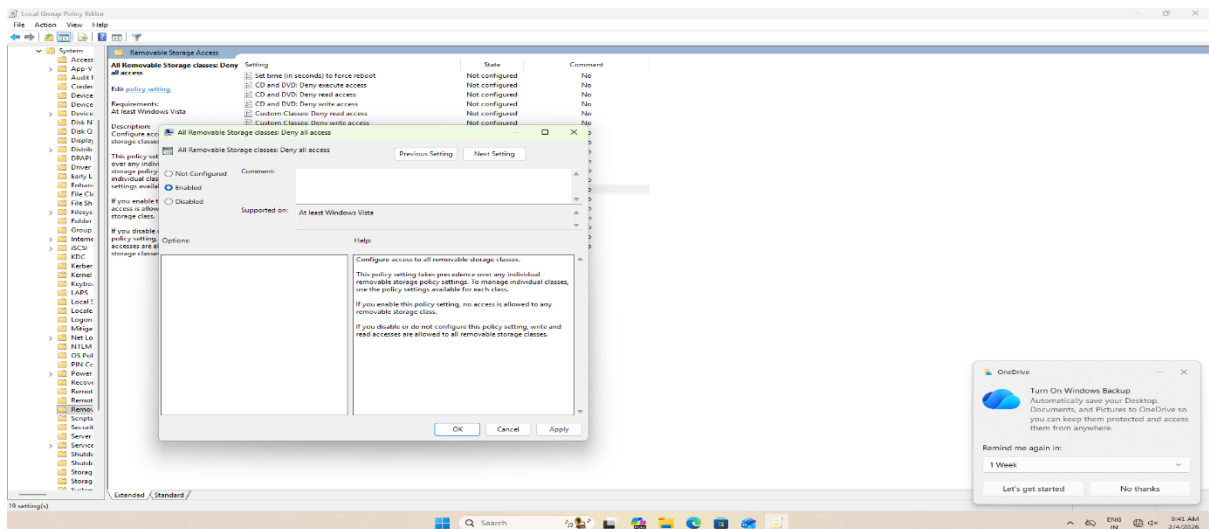
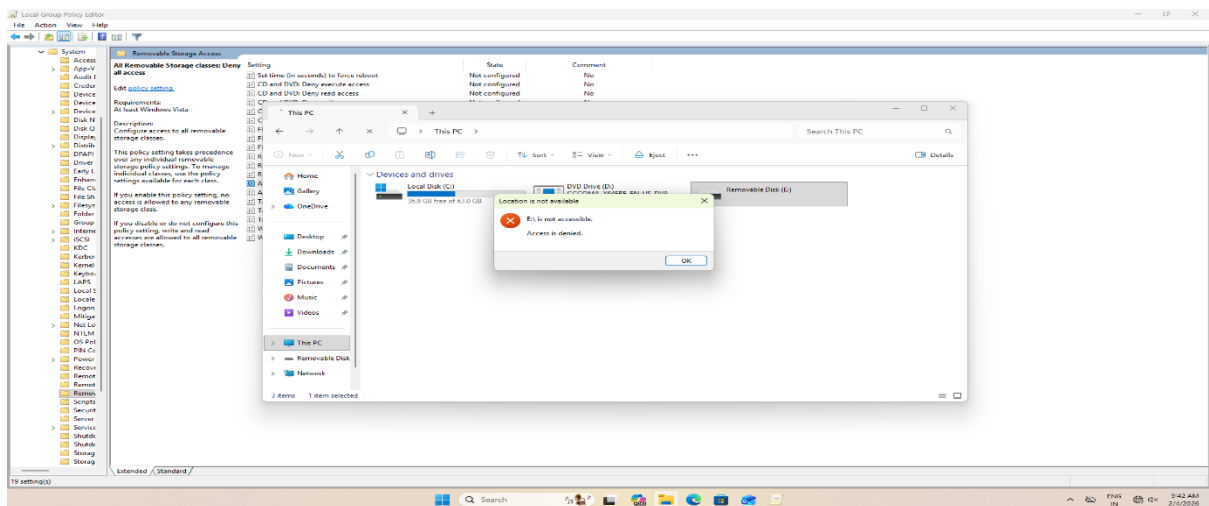


Image 9: USB Access Denied

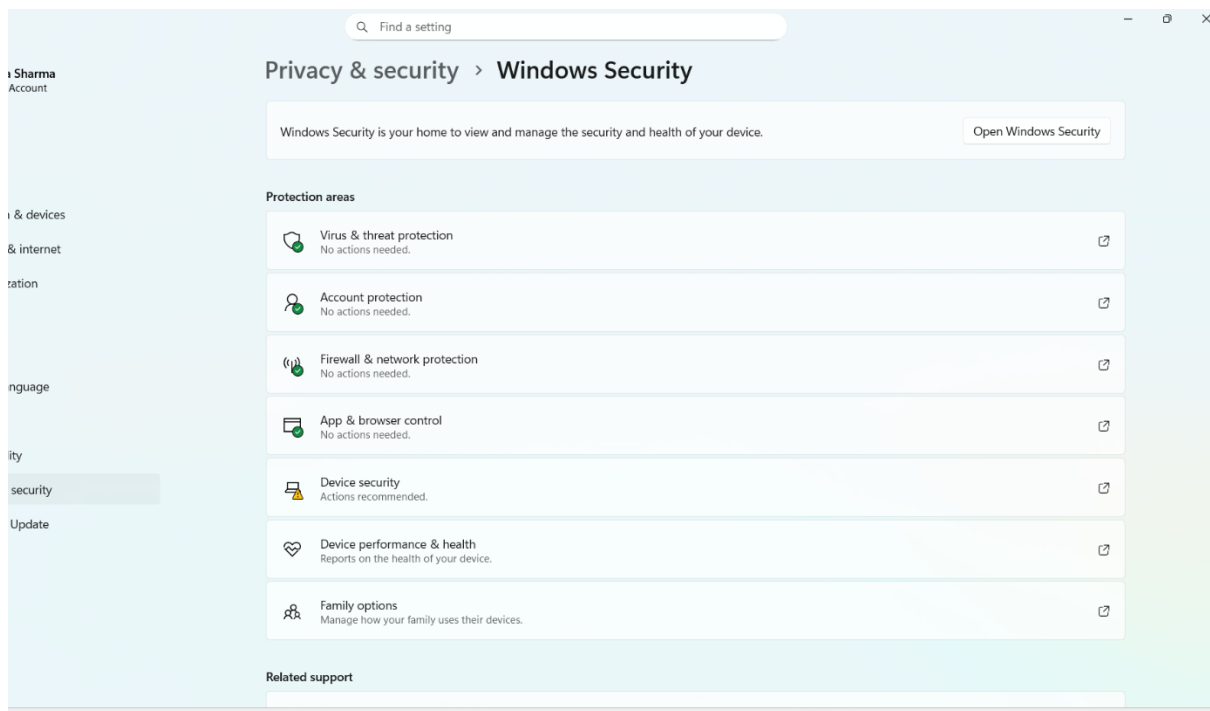


Step 7: Verify Windows Security

- Open **Windows Security Dashboard**

- Ensure **Windows Defender Antivirus** and **Firewall** are enabled

Image 10: Windows Security Dashboard



1.5 Result

The virtual machine environment was created using VMware Workstation and Windows 11 Enterprise was installed. User accounts were configured, system updates were applied, and security policies such as password protection and USB access restriction were implemented successfully. The system is now configured as a secure virtual environment.

Unit 2: Snapshot Creation for Rollback

2.1 Agenda

The purpose of this unit is to create a snapshot of the virtual machine and demonstrate how the system can be restored to a previous state when changes or issues occur.

2.2 Introduction

Snapshots are an important feature of virtualization technology. They allow administrators to capture the current state of a virtual machine and store it as a restore point.

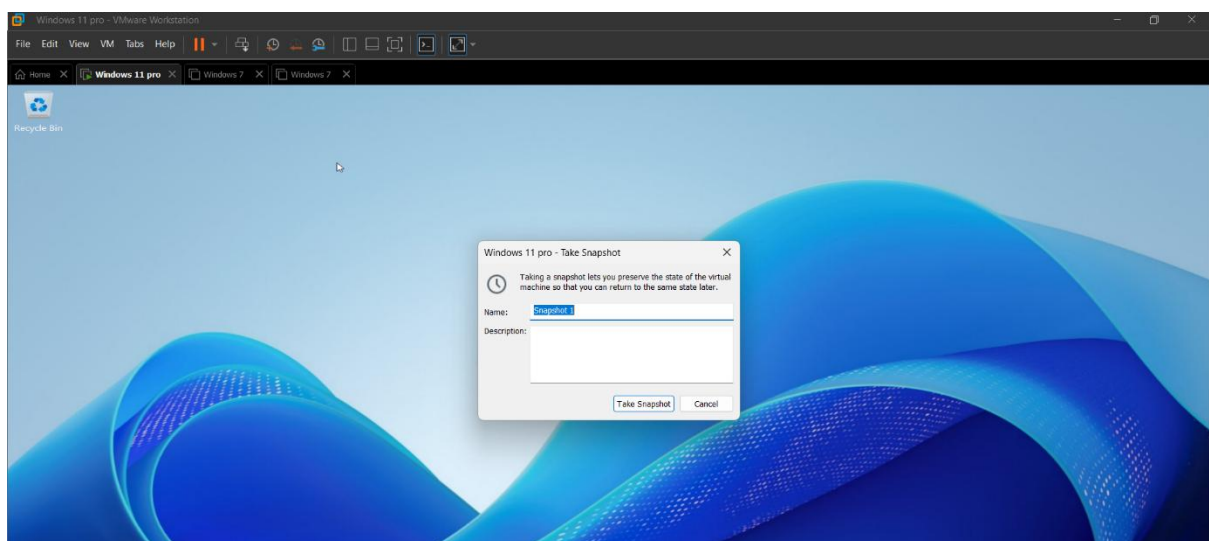
If any software installation or configuration change causes problems, the system can quickly return to the previous stable state using the snapshot.

2.3 Steps

Step 1: Create Snapshot

- Open VMware Workstation
- Select the virtual machine
- Click **VM → Snapshot → Take Snapshot**
- Enter the snapshot name **“Clean OS Before Testing”**

Image 1: Snapshot Confirmation Screen



Step 2: Make System Changes

- Install brave browser software after snapshot creation
 - Perform testing or configuration changes in the system
-

Step 3: Revert to Snapshot

- Open **Snapshot Manager**
- Select the saved snapshot
- Click **Revert to Snapshot**

The system returns to the original state before changes were made.
Installed Brave browser application is also removed

2.4 Result

The snapshot feature was successfully used to capture the system state and restore it when needed. After reverting the snapshot, the system returned to the earlier stable condition and the changes made after the snapshot were removed.

Unit 3: Installation of Packaging Tool

3.1 Agenda

The agenda of this unit is to download, install, and configure AdminStudio packaging software inside the virtual machine so that application installations can be monitored and converted into MSI packages for enterprise deployment.

3.2 Introduction

Application packaging is a process where a normal software installer is converted into a standardized Windows Installer package (.MSI). This format allows software to be deployed automatically across many computers within an organization.

AdminStudio, developed by the Flexera, is a widely used application packaging solution. It monitors system changes during software installation and creates MSI packages that can be distributed to multiple systems.

3.3 Requirements

Software

- VMware Workstation Virtual Machine
- Windows 10 / Windows 11 (Pro or Enterprise)
- Internet Connection
- AdminStudio 2025 Trial Version

Hardware

- Minimum 4 GB RAM
- Minimum 40 GB Hard Disk Space

3.4 Steps

Step 1: Visit Flexera Website

- Open a web browser inside the virtual machine
- Go to the official Flexera website
- Search for AdminStudio trial download

Step 2: Register for Trial Version

- Fill in required information such as name, email, company, and job role
- Submit the trial registration form

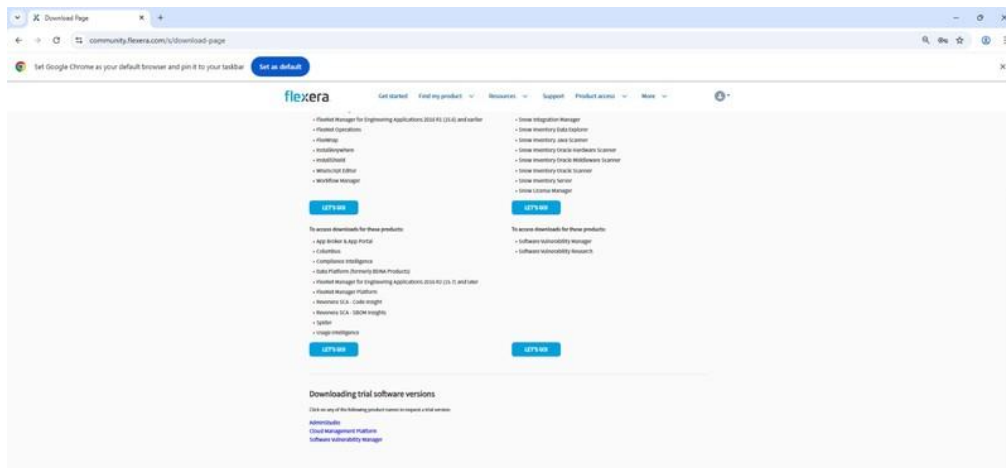
Image 2: Trial Registration Form

The screenshot shows a web browser window with the URL `info.flexera.com/AR-EVAL-AdminStudio`. The page features a blue header with the text "Try AdminStudio Enterprise FREE for 14 days!" and a "Try in Azure" button. Below the header, there is a section titled "Powering the Application Readiness Process for planning, packaging and delivery of physical, virtual, and mobile applications, ensuring faster service delivery and continuous and predictable deployment". This section includes a paragraph about AdminStudio Enterprise and a list of bullet points describing its capabilities. On the right side of the page, there is a "GET YOUR FREE TRIAL" button with a downward arrow. Below this button, there is a form with the following fields: "Work Email" (with the value `nitish.bscitstudent23.9505@cimage.in`), "First Name" (with the value `Nitish`), "Last Name" (with the value `Sharma`), "Company" (with the value `Personal`), and "Job Title" (with the value `Desktop Administrator`). A note above the form states: "Please complete this form and a Flexera representative will be in touch with you shortly."

Step 3: Access Flexera Portal

- Log in to the Flexera community portal
- Navigate to product downloads
- Locate the AdminStudio trial version

Image 3: Trial Download Page



Step 4: Download Setup File

- Download the setup file **AdminStudio2025R3.exe** from the portal

Step 5: Run the Installer

- Right-click the downloaded setup file
- Run it with administrator privileges

Step 6: Start Installation

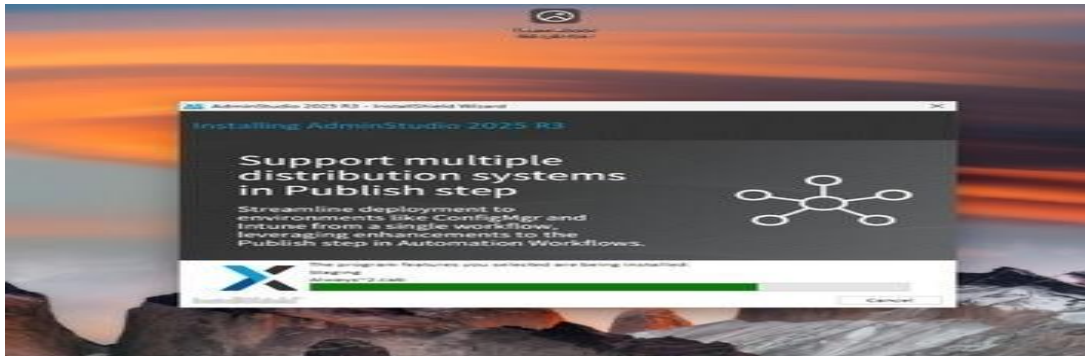
- The InstallShield installation wizard opens
- Select default installation settings

Image 6: AdminStudio Installation Wizard

Step 7: Install Components

- The installer automatically installs required components
- Wait for installation progress to complete

Image 4: Installation Progress



Step 8: Complete Installation

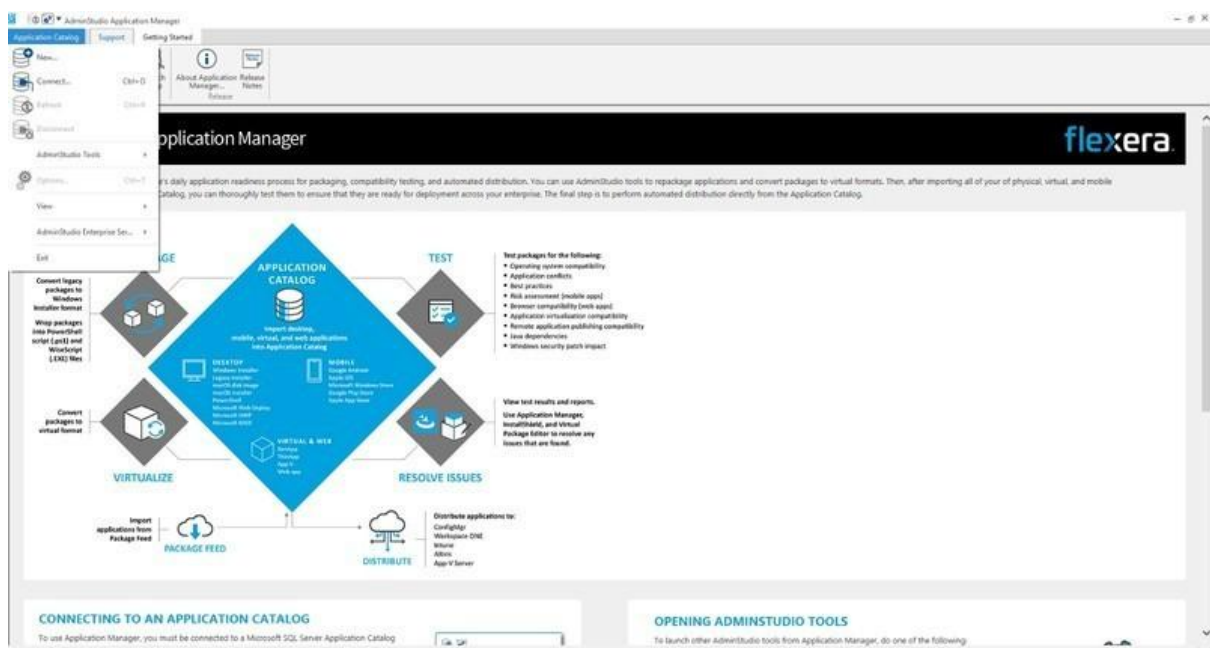
- The setup process finishes successfully
- The completion message appears on screen

Image 5: Installation Completed Screen

Step 9: Launch AdminStudio

- Open AdminStudio from the Start Menu
- The Application Manager dashboard appears

Image 6: AdminStudio Dashboard



3.5 Result

AdminStudio 2025 R3 was successfully downloaded and installed in the virtual machine. The software opened correctly and the system is now prepared for capturing application installations and generating MSI packages.

Unit 4: Capture an Application Installation Using MSI

4.1 Agenda

The agenda of this unit is to monitor the installation of a traditional software setup and convert it into a Windows Installer (.MSI) package using AdminStudio Repackager.

4.2 Introduction

Application repackaging is the process of converting standard setup files such as **.EXE installers** into **MSI packages**. MSI files are widely used in enterprise environments because they support automated software deployment.

Organizations use deployment tools like Microsoft Intune, Group Policy, or system management tools to install software across multiple computers. These tools require applications to be in MSI format.

During repackaging, AdminStudio monitors system changes such as files copied, registry entries created, and shortcuts added. After capturing these changes, it builds a new MSI installer that can reproduce the same installation.

4.3 Software Requirements

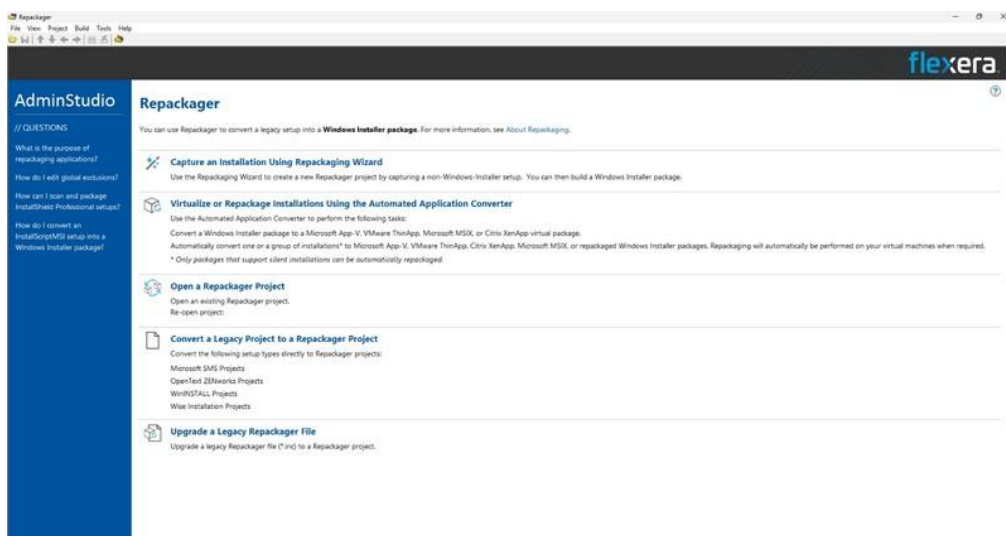
- AdminStudio Repackager
 - VMware Workstation
 - Windows 11 Virtual Machine
 - 7-Zip Setup File (7z2301-x64.exe)
-

4.4 Steps

Step 1: Open Repackager

- Launch AdminStudio
- Click **Repackager**
- Select **Capture an Installation Using Repackaging Wizard**

Image 1: Repackager Home Screen



Step 2: Start Repackaging Wizard

- Click **Next** in the wizard
- Select **Installation Monitoring (Recommended)**

Image 2: Wizard Welcome Screen



Step 3: Select Setup File

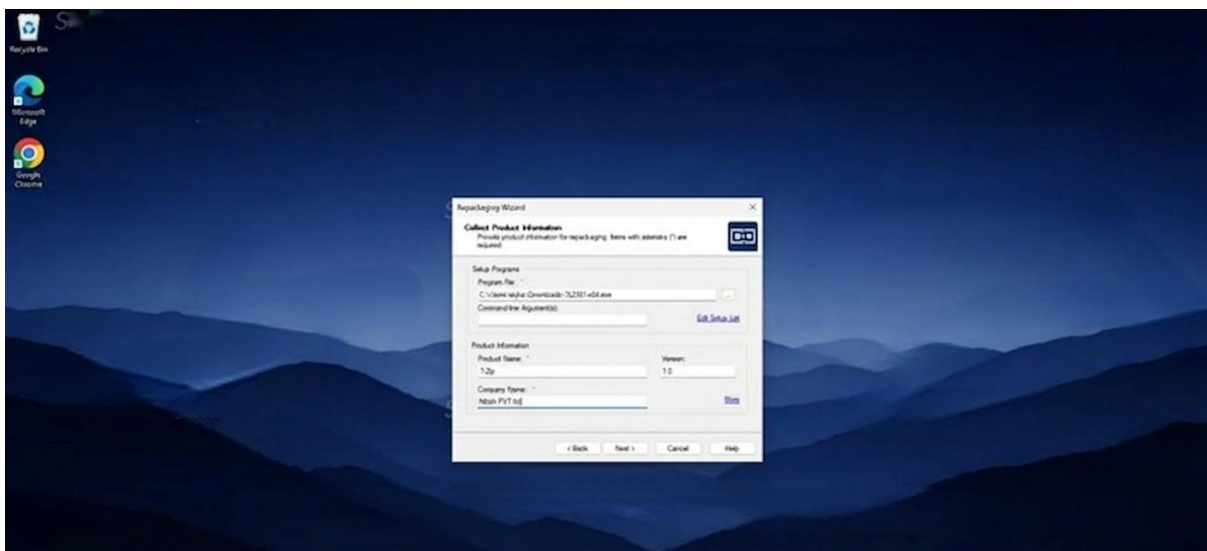
- Browse and select the installer file **7z2301-x64.exe**
- Enter product information:

Product Name: 7-Zip

Version: 1.0

Company Name: Nitish Pvt Ltd

Image 3: Product Information Screen



Step 4: Start Monitoring

- Begin monitoring process

- The system analyses current system state before installation
-

Step 5: Install Application

- The 7-Zip installer opens
- Install the application with default options
- Click **Done** after installation

Image 4: 7-Zip Installer



Step 6: Capture System Changes

AdminStudio records changes including:

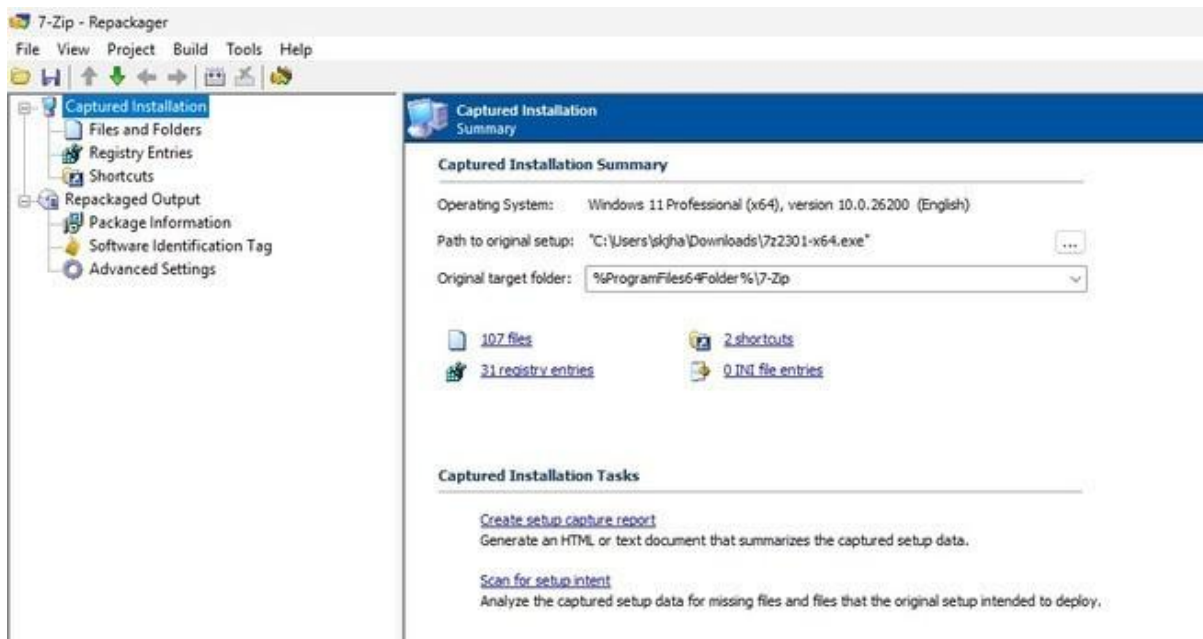
- New files created
 - Registry modifications
 - Shortcuts generated
-

Step 7: Review Captured Data

Repackager captured:

- 107 files
- 31 registry entries
- 2 shortcuts

Image 5: Files and Folders View



Step 8: Build MSI Package

- Click **Build**
- Wait for the MSI creation process to finish

Image 6: Build Log

```
Dialog SetupType for language English (United States) built
Dialog SplashBitmap for language English (United States) built
Resolving strings...
Language English (United States) built
Building .cab files...
Data1.cab built
Files built
Media table successfully built
Performing Upgrading and Patching Validation
Automatic update notification disabled
Product1\Release1 - 0 error(s), 0 warning(s)

Building software Id tag transform file...
Opening Software Id transform file
Adding files to the Software Id transform file
Adding Software Id tag 'C:\Users\skjha\RepackagerOutput\7-Zip\regid.2009-06.com.flexerasoftware.AdminStudio_7-Zip.swidtag'
Finished successfully.

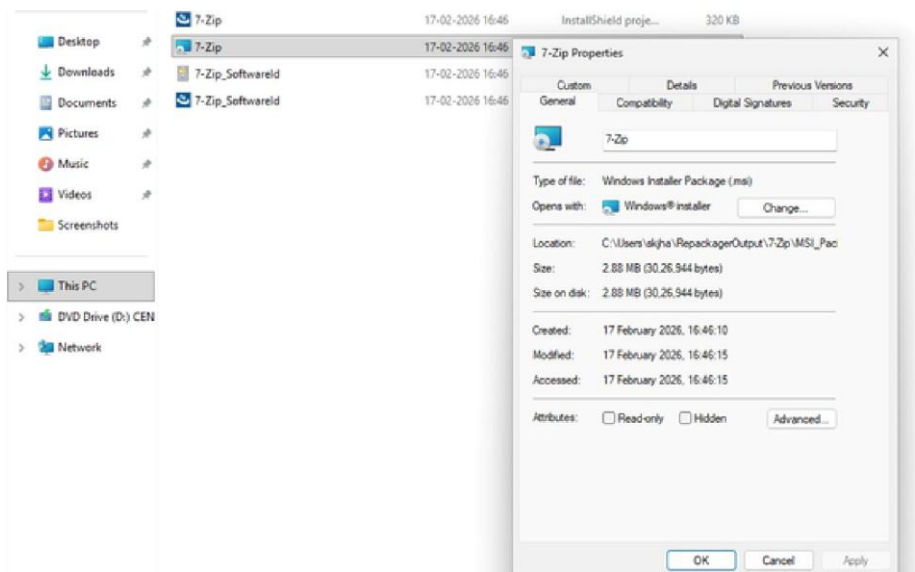
Conversion complete.
Output file:
Log file created: <file:C:\Users\skjha\RepackagerOutput\7-Zip\Build Log.txt>
```

Step 9: MSI File Creation

The MSI package is saved in:

C:\Users\...\RepackagerOutput\7-Zip\MSI_Package

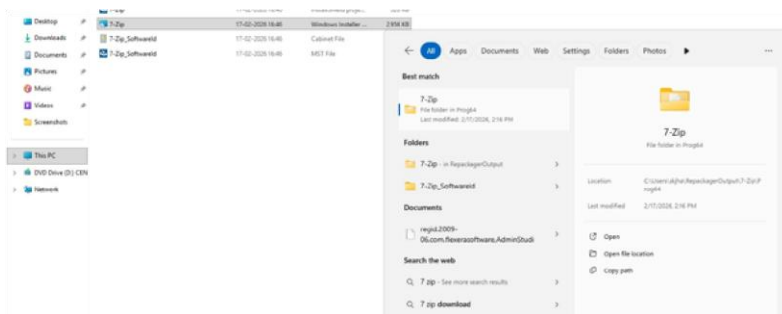
Image 7: MSI Package Folder



Step 10: Test on Clean Machine

- Revert the virtual machine to a clean snapshot
- Confirm that 7-Zip is not installed
- Run the newly created MSI package

Image 8: Clean System Before Installation



Step 11: Verify Installation

- Open Start Menu
- Search for **7-Zip File Manager**
- Launch the application

Image 9: MSI Installation Started



4.5 Result

The 7-Zip executable installer was successfully converted into a Windows Installer (.MSI) package using AdminStudio. The MSI file was tested on a clean virtual machine and the application installed correctly without errors.

Unit 5: Create a Silent Installation Script

5.1 Agenda

The agenda of this unit is to automate application installation by using a silent installation command and creating a batch script that installs software without any user interaction.

5.2 Introduction

Silent installation allows software to be installed automatically without displaying the graphical setup wizard. Instead of clicking

through installation screens, the system runs a command that installs the software silently.

MSI packages can be installed using the **msiexec** command in Windows. This command supports various parameters that allow administrators to install applications silently across multiple computers.

5.3 Requirements

- Windows 11 Virtual Machine
- AdminStudio Repackager
- Packaged MSI File (7-Zip.msi)
- Command Prompt or Batch Script

5.4 Steps

Step 1: Run Silent Installation Command

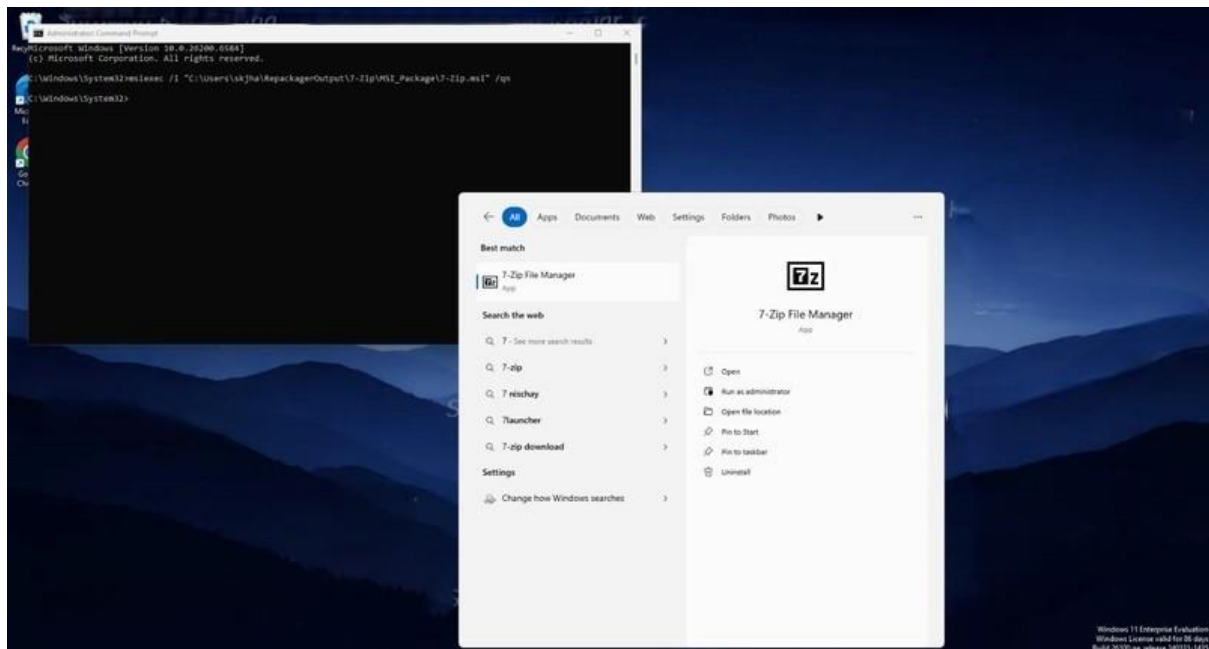
Open Command Prompt as Administrator and run:

```
"msiexec /i "C:\Users\...\7-Zip.msi" /qn"
```

Explanation:

- **msiexec** – Windows Installer service
- **/i** – Install package
- **/qn** – Quiet mode (no user interface)

Image 1: Command Prompt Running Silent Install



Step 2: Create Batch File

- Right-click Desktop
- Select **New → Text Document**
- Rename the file to **install.bat**

Add the following script:

```
@echo off
echo Installing 7-Zip silently...
msiexec /i "C:\Users\...\7-Zip.msi" /qn
echo Installation Completed
pause
```

Run the file as **Administrator**.

Image 2: install.bat File

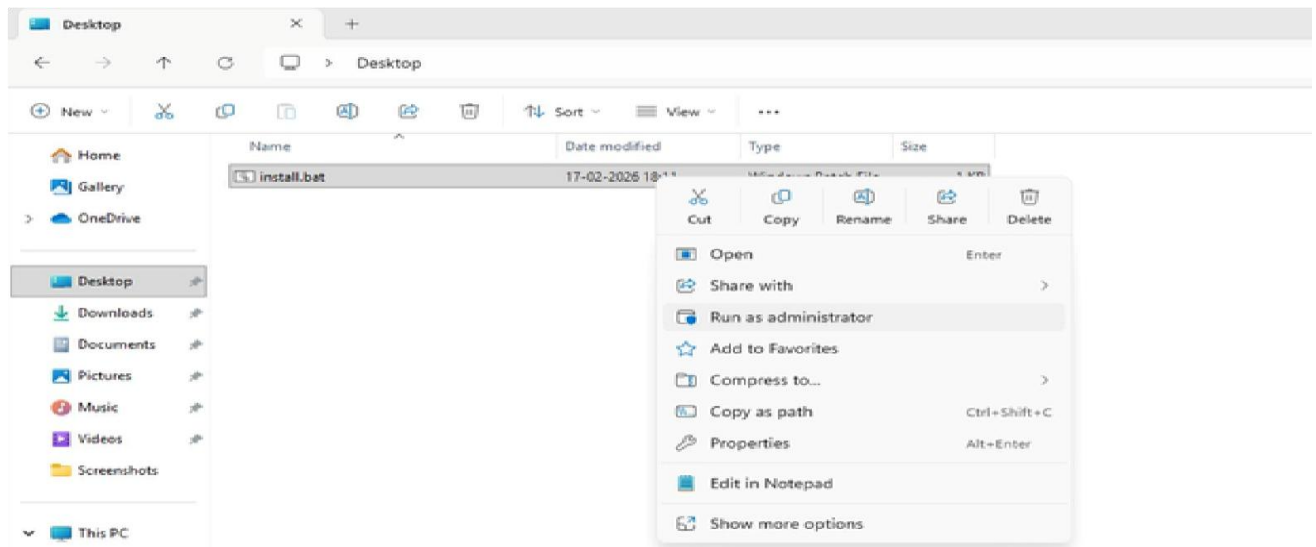
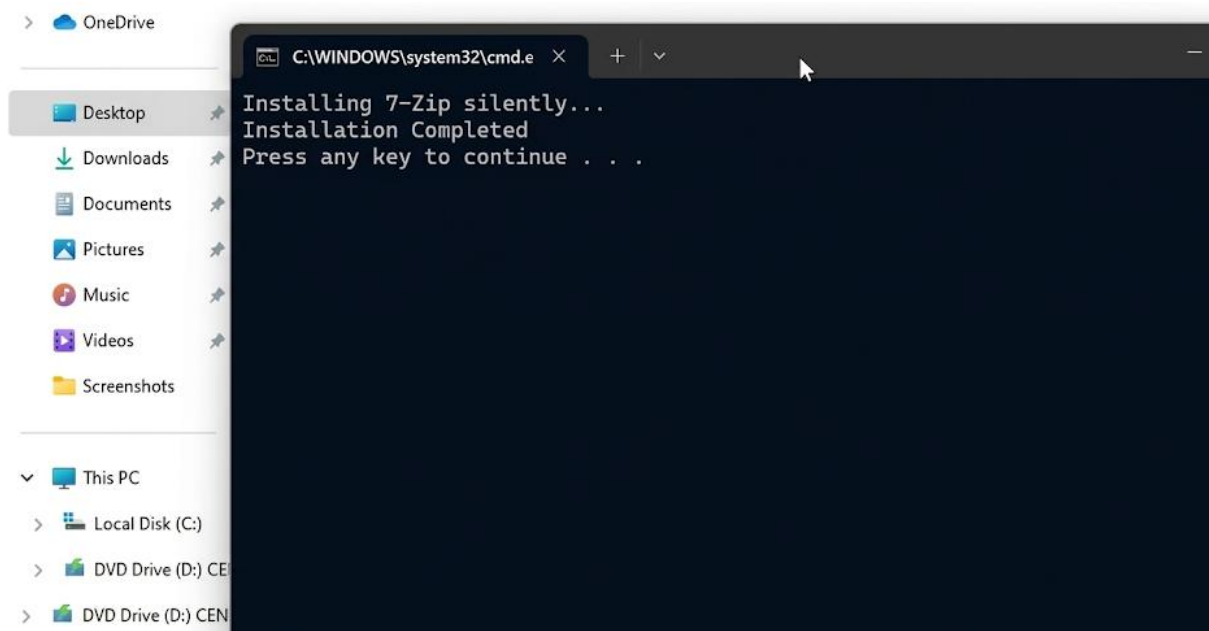


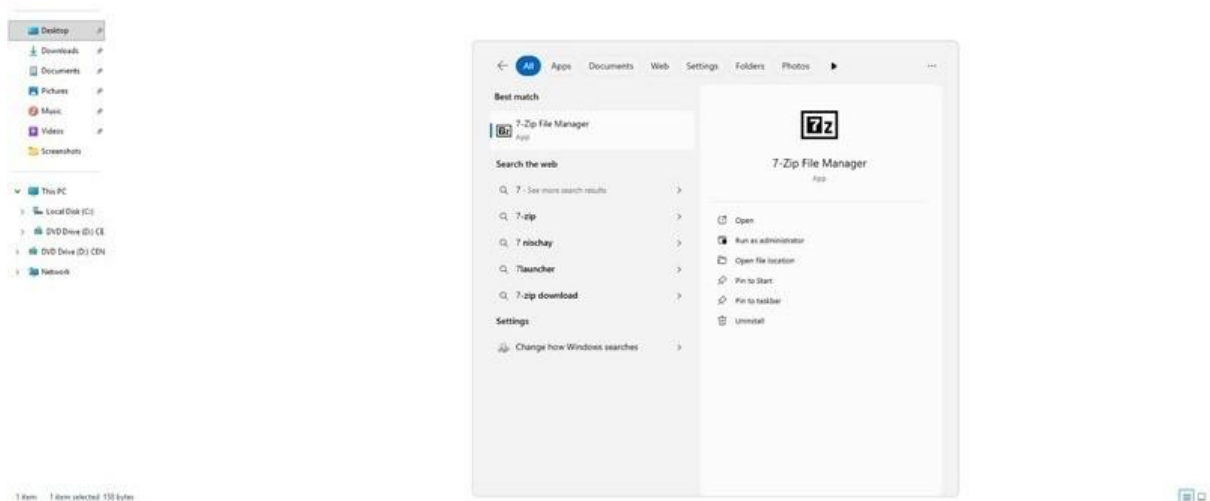
Image 3: Running Script as Administrator



Step 3: Verify Installation

- Open Start Menu
- Search **7-Zip File Manager**
- Launch the application

Image 4: 7-Zip Appears in Start Menu



5.5 Result

The 7-Zip application was successfully installed using a silent installation command and a batch script. The installation completed automatically without any user interaction.

Unit 6: Modify MSI Package Using ORCA

6.1 Agenda

The agenda of this unit is to customize the MSI package by editing its properties using the ORCA MSI database editor.

6.2 Introduction

MSI files contain a database structure that stores installation information such as product name, manufacturer, installation settings, and configuration properties. By modifying these values, administrators can control how software behaves during installation.

The ORCA tool allows users to open and edit MSI databases. It is commonly used by application packaging engineers to customize software installations according to organizational standards.

6.3 Requirements

- Windows 11 Virtual Machine
 - AdminStudio Repackaged MSI (7-Zip.msi)
 - Windows SDK
 - ORCA Tool
-

6.4 Steps

Step 1: Install Windows SDK

- Download Windows SDK from Microsoft website
- Run the setup file
- Accept license agreement
- Install with default settings

Image 1: Windows SDK Setup

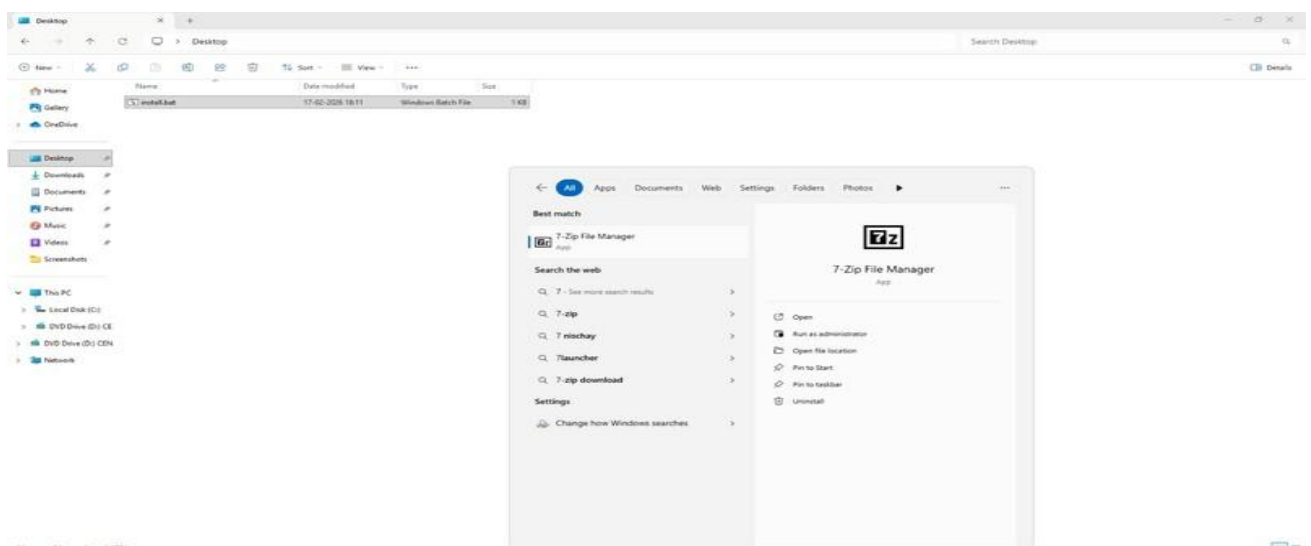
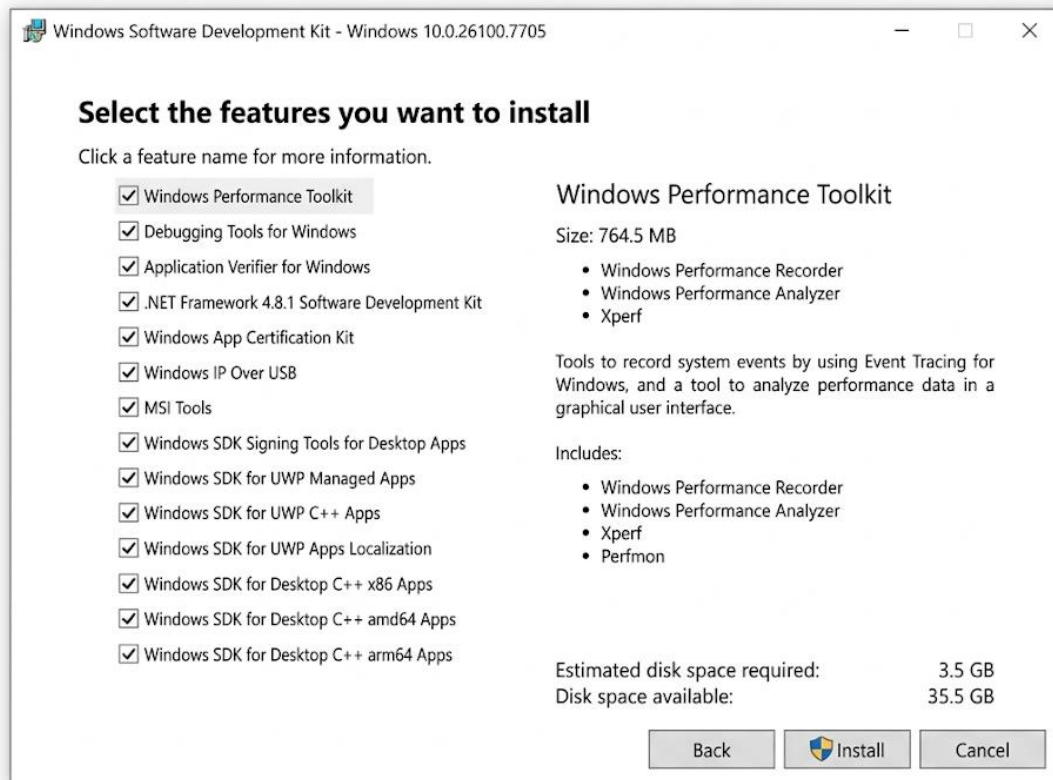


Image 2: Feature Selection



Step 2: Locate ORCA Tool

Navigate to:

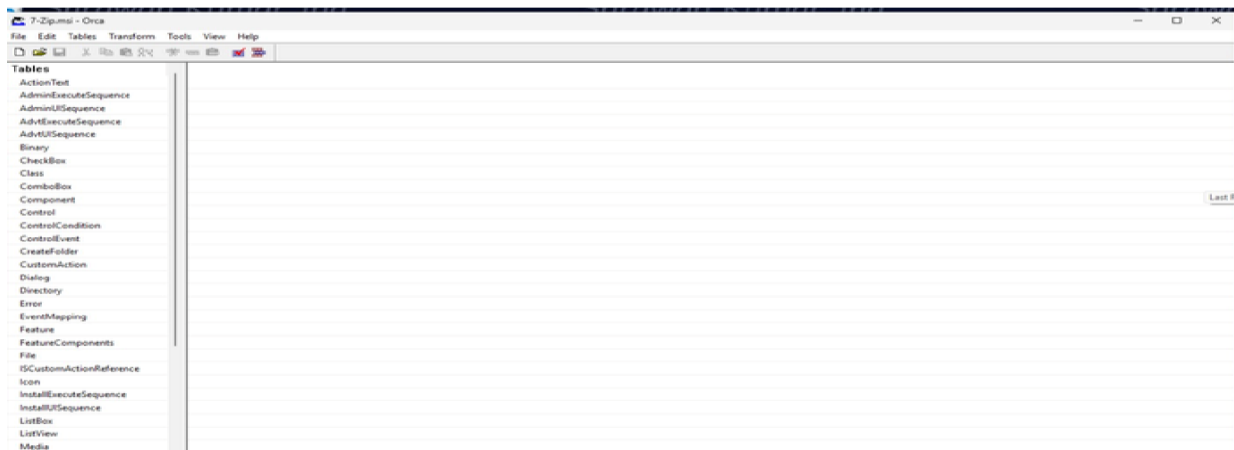
C:\Program Files (x86)\Windows Kits\10\bin\...\x86

Install **Orca-x86_en-us.msi**

Step 3: Open MSI File

- Right-click **7-Zip.msi**
- Select **Edit with Orca**

Image 3: ORCA Interface



Step 4: Modify Properties

It consists the configuration values that control how the software installs and behaves in an organization.

1. Changing Product Name

Value = 7-zip enterprise

Purpose:

To display a customized software name in company deployment environments and Programs & Features list.

2. Changing Manufacturer

Manufacturer Changed to:
Nitish Kumar Sharma

3. ALLUSERS Property

Property: ALLUSERS Value set to **1**

Meaning:

- 1 → Installed for all users (Lab/Company PC)
- 0 → Installed for only current user

Requirement:

All enterprise deployments must use ALLUSERS = 1.

1. Disabling Repair Option A new property was added:

Property	Value
ARNOREPAIR	1

2. Disabling Modify Option

Another property was added:

Property	Value
ARNOMODIFY	1

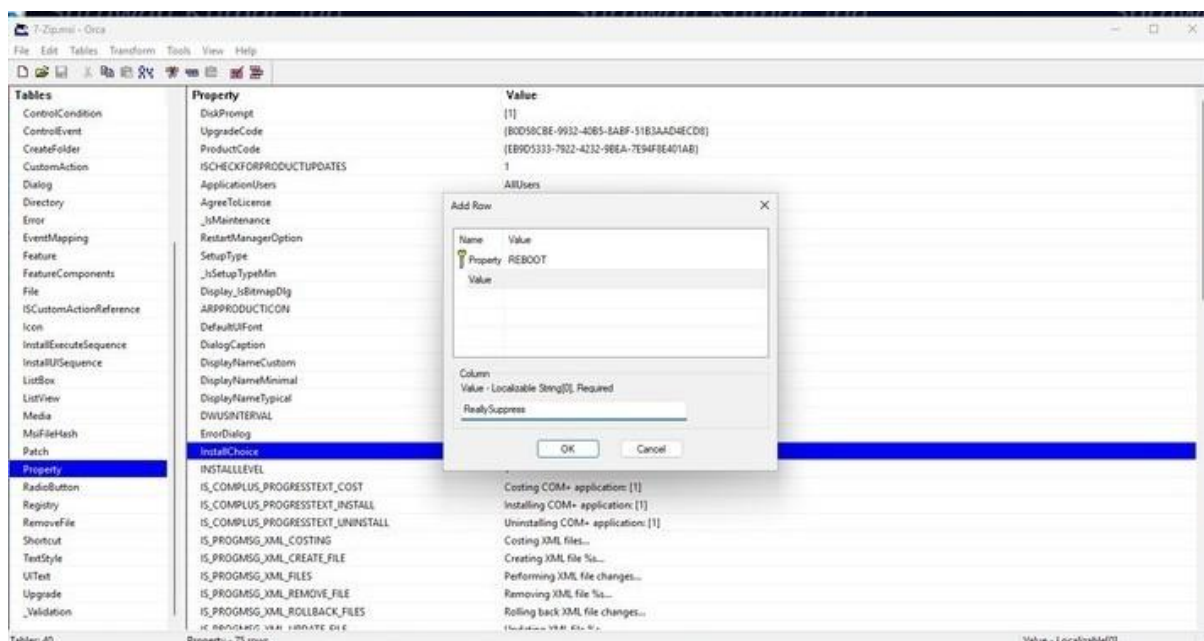
Prevents users from changing or modifying installed software.

1. Controlling System Restart

Property: REBOOT **Changed to:** ReallySuppress

This is necessary in enterprise environments because multiple applications are deployed silently.

Image 4: REBOOT property



Install Level Configuration

Property: INSTALLLEVEL

Set to: 1

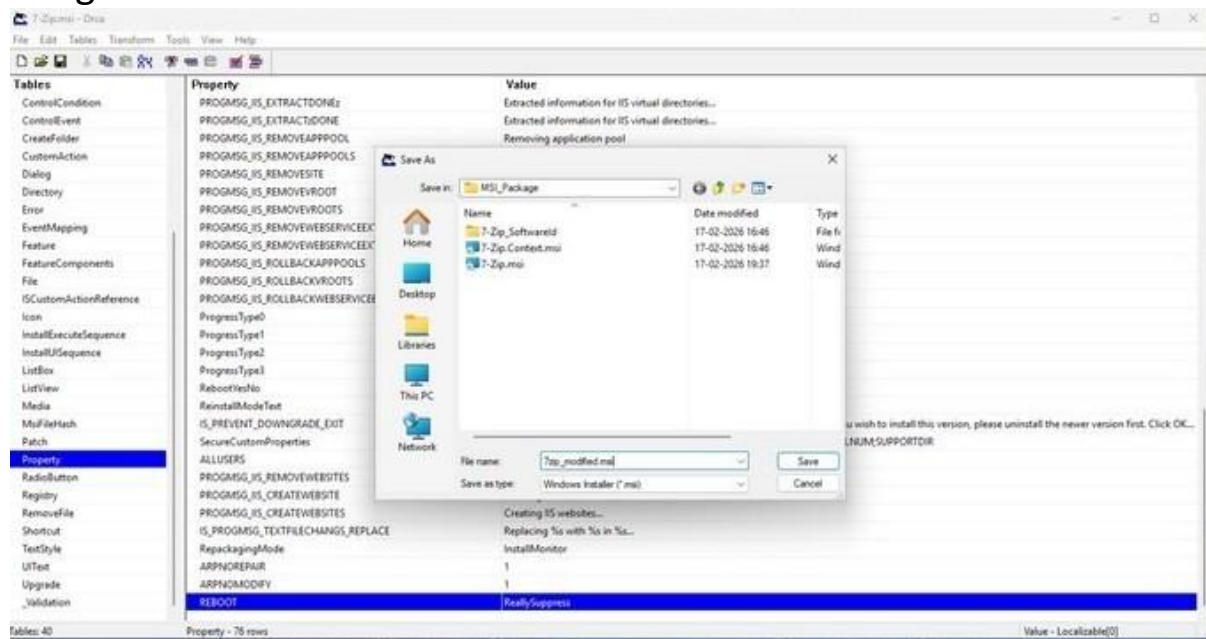
Purpose:

Ensures all application features are installed automatically without user selection.

Saving the Modified MSI After all modifications: Click File → Save As

- File saved as:
- 7zip_modified.msi

Image 5: Save as window



6.5 Result

The MSI package was successfully modified using the ORCA editor. The updated installer now installs the application for all users, prevents modification and repair options, suppresses system reboot, and displays a customized product name suitable for enterprise deployment.

Unit 7: Add Custom Registry Entries

7.1 Agenda

The agenda of this unit is to insert custom registry keys and values into a repackaged MSI installer using the ORCA MSI Editor and confirm that the registry entry is created automatically during installation.

7.2 Requirements

Software and Tools:

- Windows 10 / Windows 11 Virtual Machine
 - AdminStudio Repackager Output (7-Zip MSI)
 - ORCA MSI Editor Tool
 - Command Prompt (Administrator)
 - Modified MSI Package (7zip_modified.msi)
-

7.3 Introduction

Many applications store important configuration data inside the Windows Registry. This information may include user preferences, license information, system configuration settings, or application paths.

Instead of manually creating registry values after installation, application packaging engineers add these entries directly into the MSI installer. When the MSI package runs, the Windows Installer automatically creates the required registry entries.

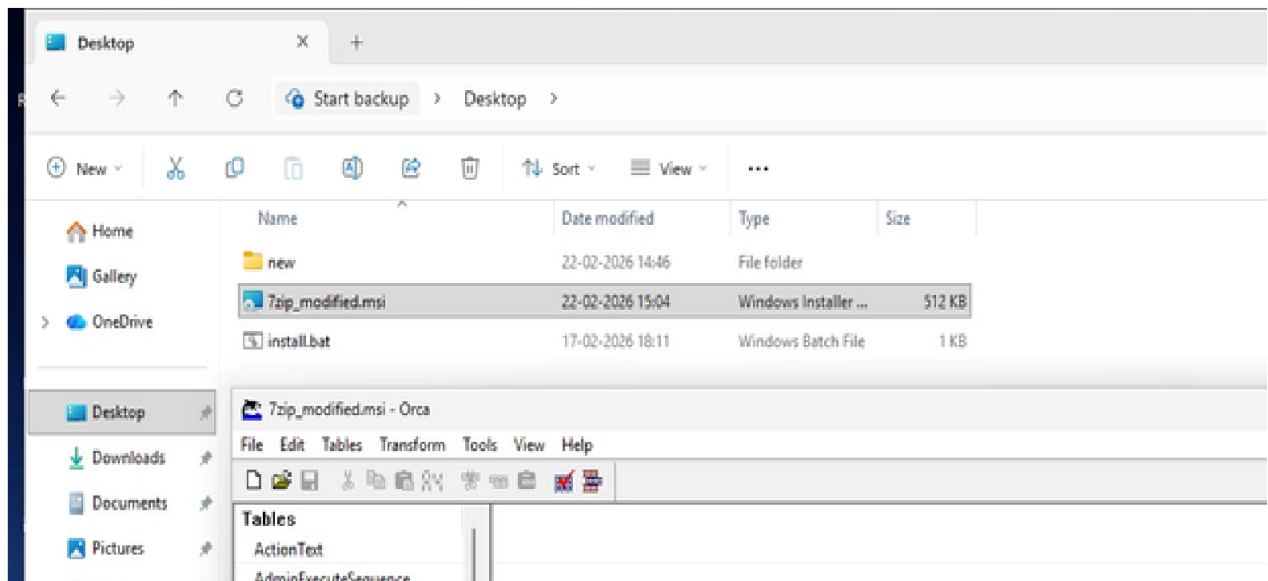
Using the ORCA tool, these entries can be added through the **Registry Table** of the MSI database.

7.4 Steps

Step 1: Open MSI File in ORCA

- Right-click **7zip_modified.msi**
- Select **Edit with Orca**

Image 1: Orca interface



Step 2: Navigate to Registry Table

- Scroll through the left panel
- Locate and select **Registry Table**

The table displays:

- Registry Root location
- Registry Key path
- Value Name
- Value Data

Image 2: Registry table entries

Tables	Registry	Fl...	Key	Name	Value	Component
ActionTest	Registry\CTD...	1	SOFTWARE\InstallShield Repackaged Applications\3d\7Zip\Folder_Created	Installed	1	Folder_Created
AdminInstallSequence	Registry\HKEY...	1	Software\7-Zip	Path	[REGTALLO...	RegistryData_MCU
AdminSequence	Registry1	2	SOFTWARE\Microsoft\Windows\CurrentVersion\App Paths\7zFM.exe	Path	[REGTALLO...	_7zfm.exe
AdminSequence	Registry2	2	SOFTWARE\Microsoft\Windows\CurrentVersion\App Paths\7zFM.exe	Path	[REGTALLO...	_7zfm.exe
AdminSequence	Registry...	0	CLSID\{2177F49-40C1-276A-1000-000100020000}\InprocServer32	ThreadingModel	Apartment	_7zip.dll; COM
Binary	Registry_3B4D0D1...	0	"\shell\ContentMenuHandlers\7-Zip		{2177F49-40C1-276A-1000-000100020000}	RegistryData_MCR64
Checkbox	Registry_C7D2B9...	0	Directory\shell\ContentMenuHandlers\7-Zip		{2177F49-40C1-276A-1000-000100020000}	RegistryData_MCR64
Class	Registry_8B8E1...	0	Directory\shell\DragDropHandlers\7-Zip		{2177F49-40C1-276A-1000-000100020000}	RegistryData_MCR64
ComboBox	Registry_F8C81...	0	Drive\shell\DragDropHandlers\7-Zip		{2177F49-40C1-276A-1000-000100020000}	RegistryData_MCR64
Component	Registry_7C86AFA...	0	Folder\shell\ContentMenuHandlers\7-Zip		{2177F49-40C1-276A-1000-000100020000}	RegistryData_MCR64
Control	Registry_62A2B8...	0	CLSID\{2177F49-40C1-276A-1000-000100020000}	7-Zip Shell Extension		RegistryData_MCR
ControlCondition	Registry_F8A2D5...	0	CLSID\{2177F49-40C1-276A-1000-000100020000}\InprocServer32	[REGTALLO...	7zip32.dll	RegistryData_MCR
ControlEvent	Registry_7158A1...	0	CLSID\{2177F49-40C1-276A-1000-000100020000}\InprocServer32	ThreadingModel	Apartment	RegistryData_MCR
CreateFolder	Registry_9F73D5...	1	Software\7-Zip	Path64	[REGTALLO...	RegistryData_MCU
CustomAction	Registry_9F73D5...	2	Software\7-Zip	Path	[REGTALLO...	RegistryData_MCU64
Dialog	Registry_2D476A...	2	Software\7-Zip	Path64	[REGTALLO...	RegistryData_MCU64
Directory	Registry_9BA1B8...	2	Software\Microsoft\Windows\CurrentVersion\Shell Extensions\Approved		{2177F49-40C1-276A-1000-000100020000}	RegistryData_MCU64
Error	Registry_FF85FA...	2	Software\Microsoft\Windows\CurrentVersion\Shell Extensions\Approved		{2177F49-40C1-276A-1000-000100020000}	RegistryData_MCU64
EventLogging						
Feature						
FeatureComponents						

Step 3: Create a Custom Registry Entry

- Click **Tables** → **Add Row**
- A new dialog window appears
- Enter the registry details manually

Custom registry values used:

Field Value

Root 2 (HKLM)

Key Software\7zipEnterprise

Name Company

Value Nitish Kumar Sharma

This configuration ensures the registry entry is created automatically during installation.

Image 3: Add Row window

- Press **Win + R**
- Type **regedit**
- Navigate to:

HKEY_LOCAL_MACHINE

→ SOFTWARE

→ 7zipEnterprise

You should see:

Company = Nitish Kumar Sharma

7.5 Result

The registry entry was successfully added into the MSI package using ORCA. During installation, the Windows Installer automatically created the registry key and value without manual configuration.

Unit 8: Package an Application with Prerequisites

8.1 Agenda

The agenda of this unit is to package an application along with its required prerequisite software so that the program installs successfully even on systems that do not already have required components.

8.2 Introduction

Some applications cannot run properly without additional components called **prerequisites**. These supporting libraries must exist on the system before the main application can run.

Common prerequisites include:

- Microsoft .NET Framework

- Microsoft Visual C++ Redistributable
- Java Runtime Environment

If these components are missing, the application may show errors such as:

- Missing DLL files
- Application startup failure
- Runtime errors

To avoid these problems, packaging engineers include prerequisite installers inside the application package.

In this experiment, the prerequisite **Microsoft Visual C++ 2013 Redistributable** was packaged using AdminStudio Repackager.

8.3 Requirements

- Windows 11 Virtual Machine
 - AdminStudio Repackager
 - Visual C++ Redistributable Installer (vc_redist.x64.exe)
 - Clean Snapshot System
-

8.4 Steps

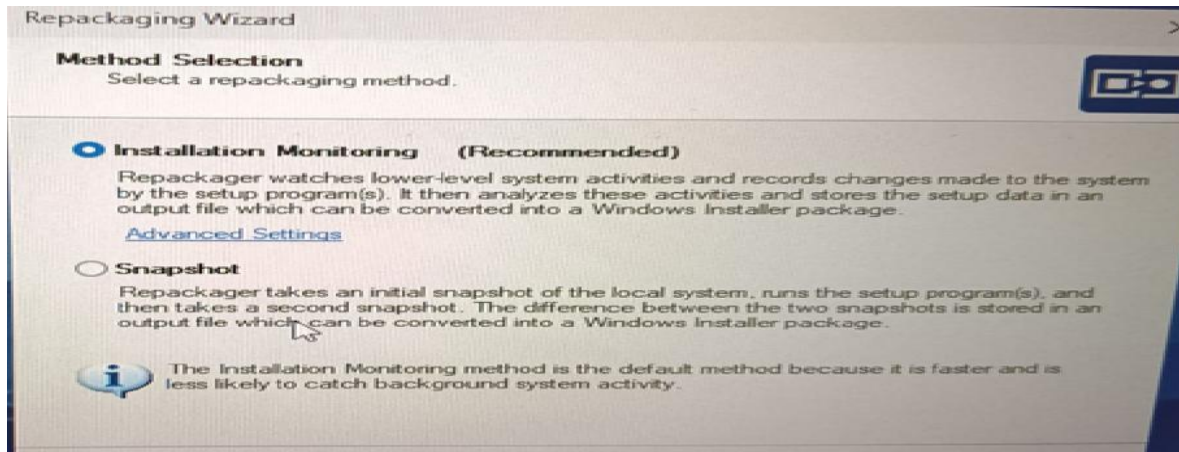
Step 1: Prepare Clean Environment

- Revert the virtual machine to a clean snapshot
 - Close unnecessary background applications
 - Ensure no prerequisite software is already installed
-

Step 2: Launch Repackager Wizard

- Open AdminStudio Repackager
- Select **Capture an Installation Using Repackaging Wizard**
- Choose **Installation Monitoring (Recommended)**

Image 1: Repackaging method selection



Step 3: Select Setup File

- Browse and select: vc_redist.x64.exe

Enter product details:

Product Name: Microsoft Visual C++ 2013 Redistributable (x64)

Company Name: Microsoft Corporation

Version: 1.0

Step 4: Start System Capture

- Click **Start Monitoring**
- Repackager captures the system snapshot before installation

Step 5: Install Prerequisite

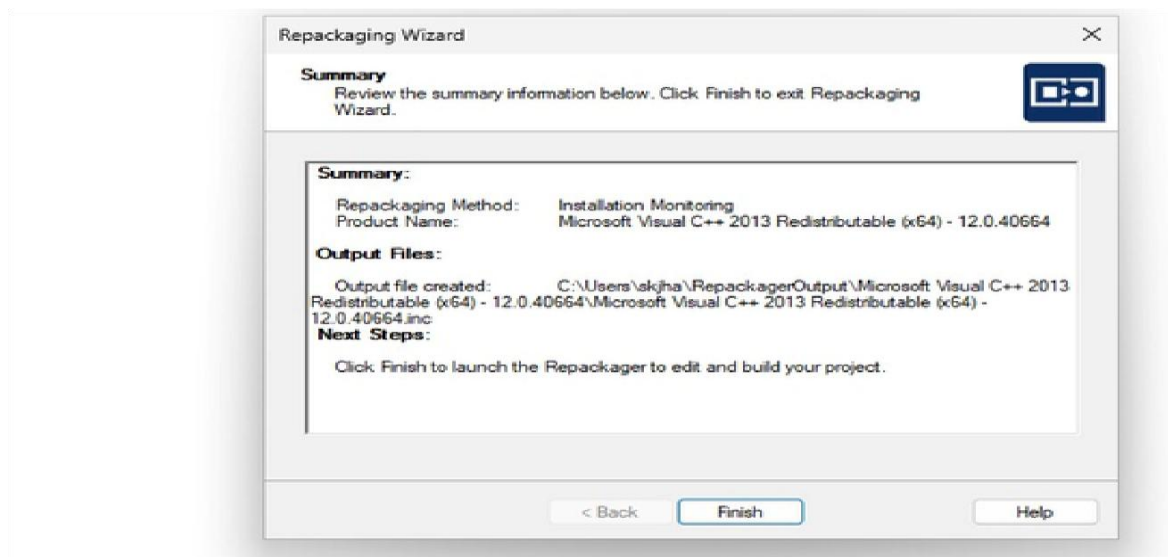
- The Visual C++ installer opens

- Click **Install**
 - Wait for installation to finish
 - Do not restart the system
-

Step 6: Stop Monitoring

- Return to Repackager
- Click **Stop Monitoring**
- The tool compares system changes

Image 2: Captured installation summary



Step 7: Build MSI Package

- Click **Build → Build MSI**
- Wait until conversion completes

Output location:

C:\Users\...\RepackagerOutput\
Microsoft Visual C++ 2013 Redistributable (x64)

Image 3: Build log showing completion

```
Dialog MaintenanceType for language English (United States) built
Dialog MaintenanceWelcome for language English (United States) built
Dialog MsiRMFilesInUse for language English (United States) built
Dialog OutOfSpace for language English (United States) built
Dialog PatchWelcome for language English (United States) built
Dialog ReadyToInstall for language English (United States) built
Dialog ReadyToRemove for language English (United States) built
Dialog SetupCompleteError for language English (United States) built
Dialog SetupCompleteSuccess for language English (United States) built
Dialog SetupError for language English (United States) built
Dialog SetupInitialization for language English (United States) built
Dialog SetupInterrupted for language English (United States) built
Dialog SetupProgress for language English (United States) built
Dialog SetupResume for language English (United States) built
Dialog SetupType for language English (United States) built
Dialog SplashBitmap for language English (United States) built
Resolving strings...
Language English (United States) built
Building .cab files...
Data1.cab built
Files built
Media table successfully built
Performing Upgrading and Patching Validation
Automatic update notification disabled
Product1\Release1 - 0 error(s), 0 warning(s)

Building software Id tag transform file...
Opening Software Id transform file
Adding files to the Software Id transform file
Adding Software Id tag 'C:\Users\skjha\RepackagerOutput\Microsoft Visual C++ 2013 Redistributable (x64)
Finished successfully.

Conversion complete.
Output files:
Log file created: <file:C:\Users\skjha\RepackagerOutput\Microsoft Visual C++ 2013 Redistributable (x64) .
```

8.5 Result

The prerequisite package **Microsoft Visual C++ 2013 Redistributable (x64)** was successfully captured and converted into an MSI package using AdminStudio Repackager.

Unit 9: Create a Transform (.MST) File

9.1 Agenda

The agenda of this unit is to customize an MSI installer by creating a Transform (.MST) file using the ORCA tool without modifying the original MSI package.

9.2 Tools Used

- AdminStudio Repackager
- ORCA MSI Editor
- Command Prompt (Administrator)
- Windows 11 Virtual Machine

- 7-Zip MSI Installer

9.3 Introduction

In enterprise environments, administrators avoid editing the original MSI installer because doing so can break the digital signature or affect vendor support.

Instead, they create a **Transform (.MST)** file that stores only the changes made to the installer.

During installation:

Original MSI + MST Transform = Customized Installation

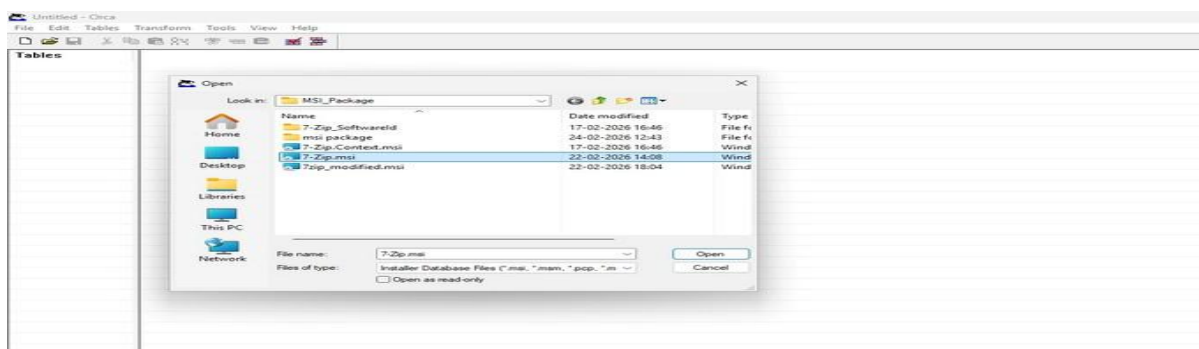
This method is widely used in enterprise deployment systems such as **Microsoft System Centre Configuration Manager**.

9.4 Steps

Step 1: Open MSI File

- Launch ORCA
- Click **File → Open**
- Select **7-Zip.msi**

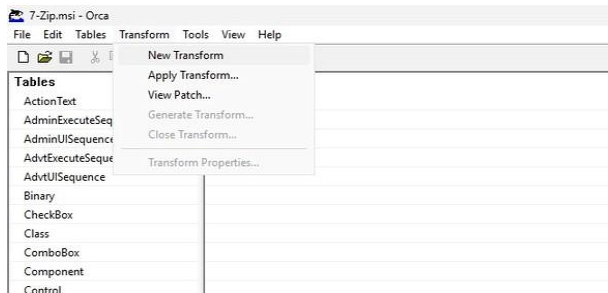
Image 1: Orca file selection window



Step 2: Create New Transform

- Click **Transform** → **New Transform**

Image 2: Orca transform menu

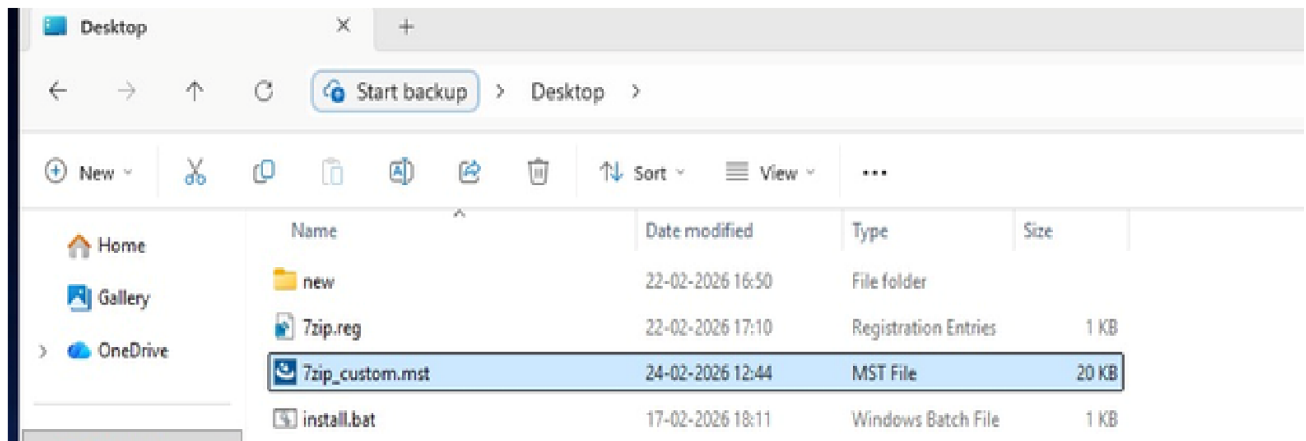


Step 3: Modify MSI Properties

Edit the Property Table values:

Property	Value
Manufacturer	Nitish Kumar Sharma
ProductName	7-Zip Enterprise
ALLUSERS	1
ARPNOREPAIR	1

Image 3: Property table modifications

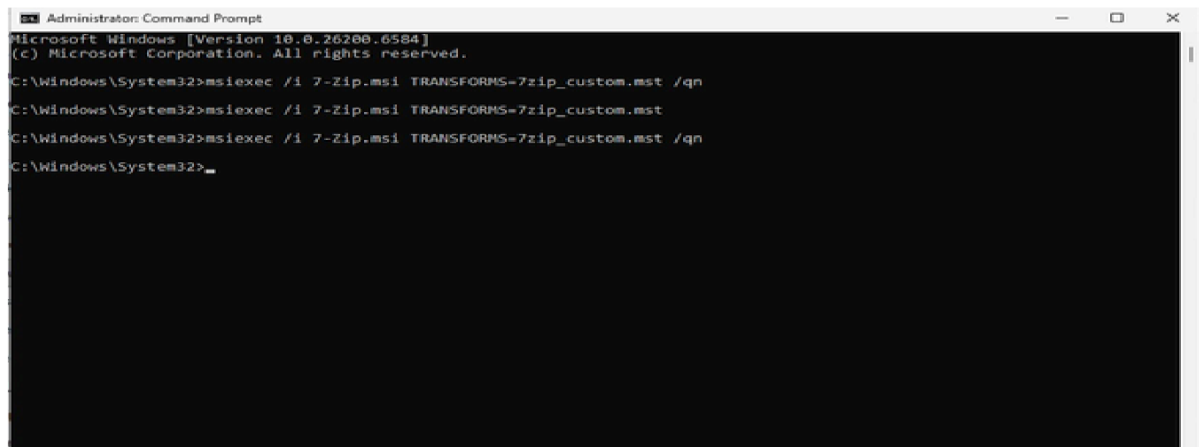


Step 5: Apply Transform During Installation

Run command in Command Prompt:

Run: `msiexec /i 7-Zip.msi TRANSFORMS=7zip_custom.mst /qn`

Image 5: Command execution

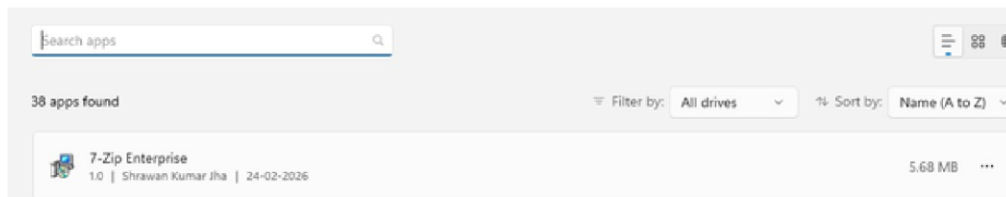


Step 6: Verify Installation

Check installed applications.

The system displays: 7-Zip Enterprise
Manufacturer: Nitish Kumar Sharma

Image 6: Installed application verification



9.5 Result

The MSI installer was successfully customized using a Transform file. The installation applied new settings without modifying the original MSI package.

Unit 10: Implement Version Control

10.1 Agenda

The agenda of this unit is to implement version control in the MSI package so that new software updates can replace the older version automatically.

10.2 Tools Used

- AdminStudio Repackager
 - ORCA MSI Editor
 - Windows 11 Virtual Machine
 - 7-Zip Packaged Application
-

10.3 Introduction

In enterprise environments, software packages must support upgrades. When a new version of software is released, the system should detect the update and replace the older installation.

To make this possible, packaging engineers modify two important MSI properties:

- **ProductVersion**
- **ProductCode (GUID)**

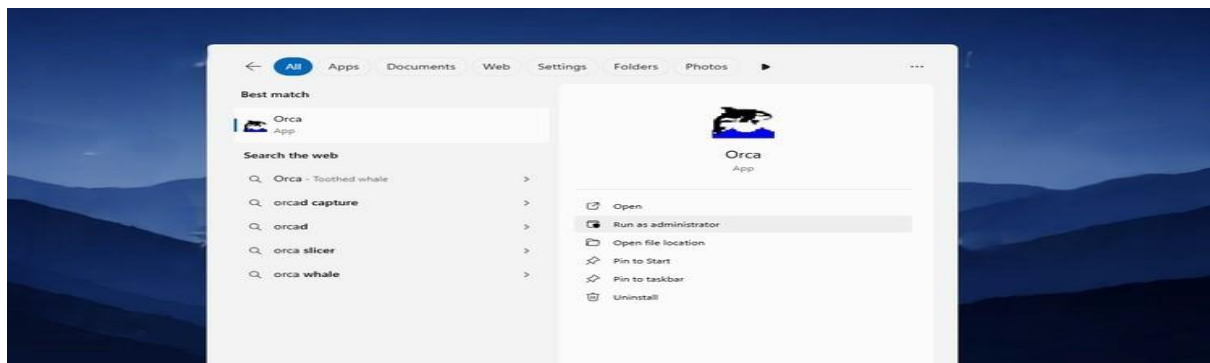
Updating these values ensures that Windows recognizes the package as a newer release.

10.4 Steps

Step 1: Open MSI in ORCA

- Launch ORCA
- Click **File → Open**
- Select **7-Zip.msi**

Image 1: Orca interface



Step 2: Edit Property Table

Locate the **Property Table** in the left panel.

Step 3: Change Product Version

Update the version number:

1.0 $\rightarrow$ 2.0

Image 2: Version modification

7-Zip.msi - Orca

File Edit Tables Transform Tools View Help

Tables

- ActionText
- AdminExecuteSequence
- AdminUISequence
- AdvtExecuteSequence
- AdvtUISequence
- Binary
- CheckBox
- Class
- ComboBox
- Component
- Control
- ControlCondition
- ControlEvent
- CreateFolder
- CustomAction

Property

Property	Value
IS_PROGMSG_XML_REMOVE_FILE	Removing XML file %s...
IS_PROGMSG_XML_ROLLBACK_FILES	Rolling back XML file changes...
IS_PROGMSG_XML_UPDATE_FILE	Updating XML file %s...
IS_SQLSERVER_AUTHENTICATION	0
IS_SQLSERVER_USERNAME	sa
ISVROOT_PORT_NO	0
ARNOREPAIR	1
Manufacturer	Shrawan Kumar Jha
PIDTemplate	12345<###- %%%%%%%%%> @@@@
ProductID	none
ProductLanguage	1033
ProductName	7-Zip Enterprise
ProductVersion	1.0
PROGMSG_IIS_CREATEAPPOOL	Creating application pool %s
PROGMSG_IIS_CREATEAPPOOLS	Creating application Pools...

Step 4: Update ProductCode

Replace the old GUID with a new GUID.

Example:

Old ProductCode

```
{EB9D5333-7922-4232-9BEA-7E94F8E401AB}
```

New ProductCode

{D6F74175-742F-41AE-9343-2AAE75BD9954}

Image 3: ProductCode change



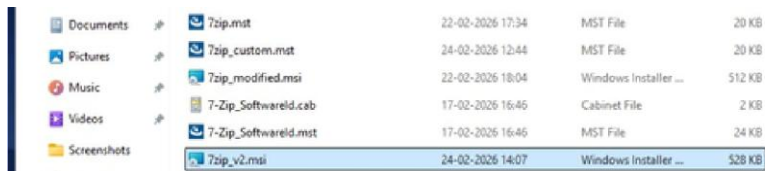
The screenshot shows the Orca application window with the file 7-Zip.msi open. The 'Tables' pane on the left lists various tables, and the 'Property' pane on the right shows the details for the 'ProductCode' property, which is highlighted in yellow. The value of the property is '{d6f74175-742f-41ae-9343-2aae75bd9954}'.

Tables	Property	Value
ActionText	DiskPrompt	{1}
AdminExecuteSequence	UpgradeCode	{B0D58CBE-9932-40B5-8ABF-51B3AAD4ECD8}
AdminUISequence	ProductCode	{d6f74175-742f-41ae-9343-2aae75bd9954}
AdvtExecuteSequence	ISCHECKFORPRODUCTUPDATES	1
AdvtUISequence	ApplicationUsers	AllUsers
Binary	AgreeToLicense	No
CheckBox	_IsMaintenance	Change
Class	RestartManagerOption	CloseRestart
ComboBox	SetupType	Typical
Component	_IsSetupTypeMin	Typical

Step 5: Save Updated MSI

Save the file as: 7zip_v2.msi

Image 4: MSI output folder



Documents	7zip.mst	22-02-2026 17:34	MST File	20 KB
Pictures	7zip_custom.mst	24-02-2026 12:44	MST File	20 KB
Music	7zip_modified.msi	22-02-2026 18:04	Windows Installer ...	512 KB
Videos	7-Zip_SoftwareId.cab	17-02-2026 16:46	Cabinet File	2 KB
Screenshots	7-Zip_SoftwareId.mst	17-02-2026 16:46	MST File	24 KB
	7zip_v2.msi	24-02-2026 14:07	Windows Installer ...	528 KB

Step 6: Install Updated Version

Run the new MSI installer.

Windows automatically detects it as a newer version and upgrades the existing application.

Image 5: Old image which is showing v1.0



Image 6: New version v2.0



10.5 Result

The application version was successfully updated from **1.0 to 2.0**. Windows Installer recognized the package as an upgrade and replaced the previous installation.

Unit 11: Test Package on Clean Virtual Machine

11.1 Agenda

The agenda of this unit is to test the packaged MSI installer on a clean virtual machine and confirm that the application installs correctly without errors.

11.2 Requirements

- Clean Windows Virtual Machine
 - Packaged MSI Application (7-Zip Enterprise v2.0)
 - AdminStudio / ORCA Tools
-

11.3 Introduction

Before deploying software in enterprise networks, it is essential to test the installation on a clean environment. This ensures that the installer works properly and does not depend on files already present in the system.

Testing on a clean virtual machine simulates a real user environment.

11.4 Steps

Step 1: Start Clean Virtual Machine

- Boot a fresh Windows virtual machine
 - Ensure that no previous version of the application exists
-

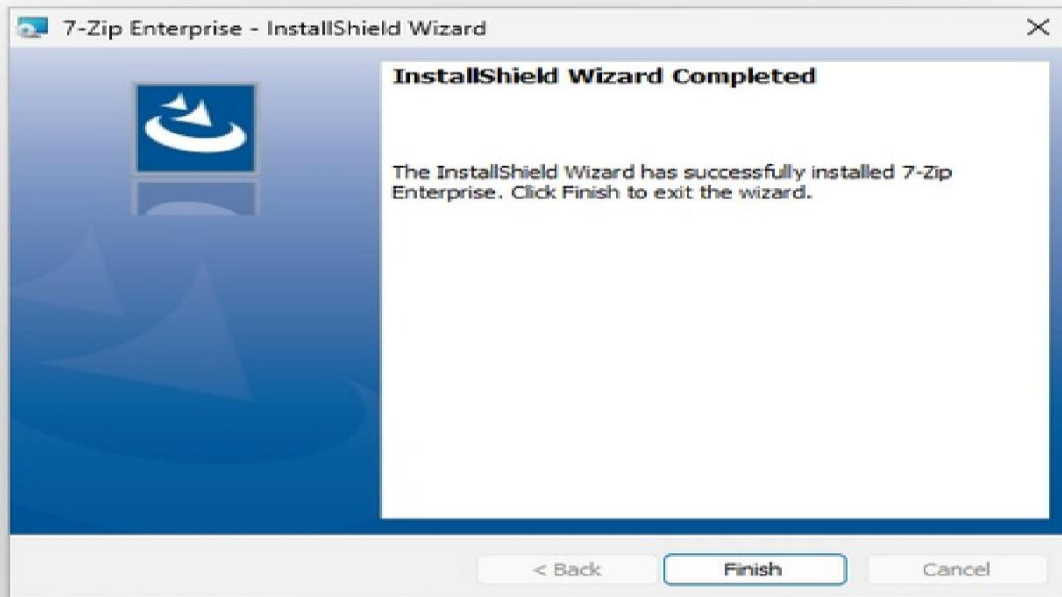
Step 2: Run the MSI Installer

- Execute **7-Zip Enterprise v2.0 MSI**
 - Installation wizard appears
-

Step 3: Complete Installation

- Click **Next**
- Accept default options
- Finish installation

Image 2: Installation completed



Step 4: Verify Application

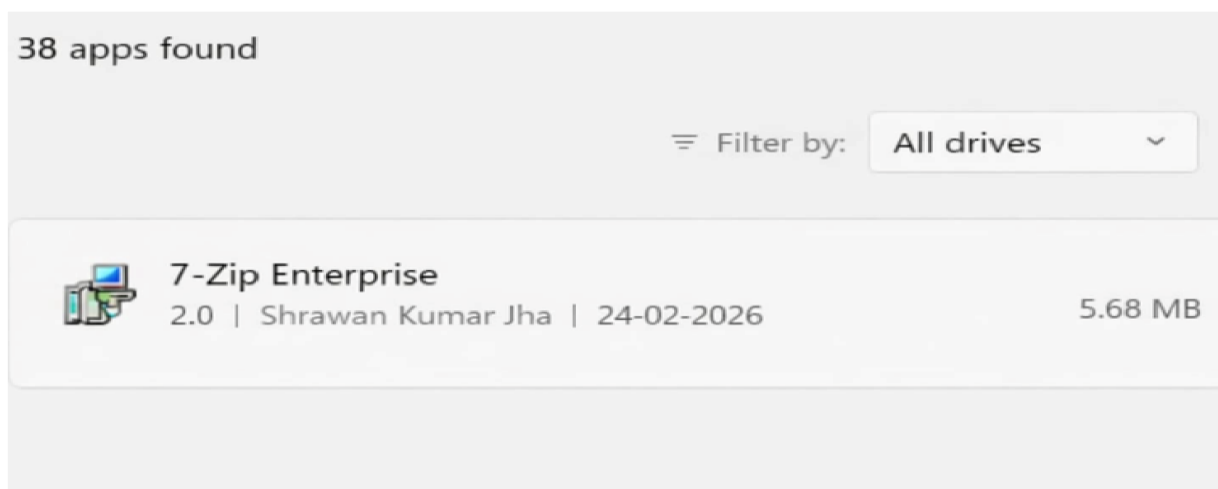
Check installed programs.

7-Zip Enterprise

Version 2.0

Manufacturer: Nitish Kumar Sharma

Image 3: Installed programs list



11.5 Result

The packaged application installed successfully on a clean virtual machine without any errors.

Unit 12: Create Uninstall Script

12.1 Agenda

The agenda of this unit is to create an automated uninstall script that removes the packaged application silently from the system.

12.2 Introduction

In enterprise environments, software removal is as important as installation. Administrators must ensure that applications can be removed completely when they are no longer required.

The Windows Installer utility **msiexec** provides command-line options that allow administrators to uninstall applications automatically.

12.3 Requirements

- Windows Virtual Machine
 - Packaged MSI Application
 - ORCA Tool
 - Command Prompt (Administrator)
-

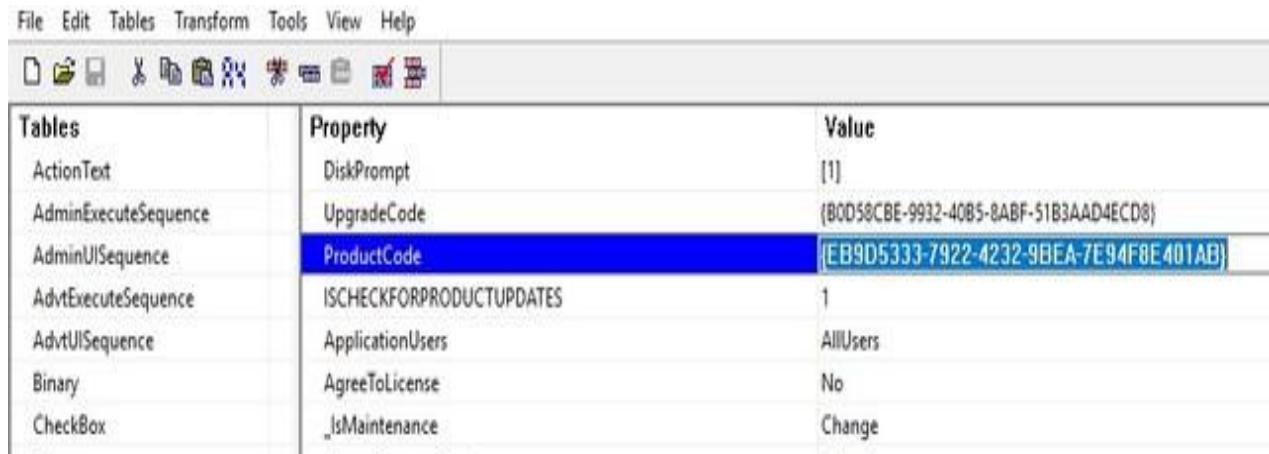
12.4 Steps

Step 1: Find Product Code

- Open MSI file in ORCA

- Navigate to **Property Table**
- Copy the **ProductCode**

Image 1: ProductCode in ORCA from Orca Property Table



The screenshot shows the Orca Property Table interface. The 'Tables' pane on the left lists various properties. The 'Property' pane in the center shows a list of properties, with 'ProductCode' selected and highlighted in blue. The 'Value' pane on the right shows the corresponding value for 'ProductCode', which is a GUID: {EB9D5333-7922-4232-9BEA-7E94F8E401AB}.

Tables	Property	Value
ActionText	DiskPrompt	[1]
AdminExecuteSequence	UpgradeCode	{B0D58CBE-9932-40B5-8ABF-51B3AAD4ECD8}
AdminUISequence	ProductCode	{EB9D5333-7922-4232-9BEA-7E94F8E401AB}
AdvtExecuteSequence	ISCHECKFORPRODUCTUPDATES	1
AdvtUISequence	ApplicationUsers	AllUsers
Binary	AgreeToLicense	No
CheckBox	_IsMaintenance	Change

Step 2: Create Uninstall Command

Use the Windows Installer command:

```
msiexec /x {ProductCode}
```

Explanation:

/x → uninstall application

{GUID} → identifies installed software

Step 3: Create Batch Script

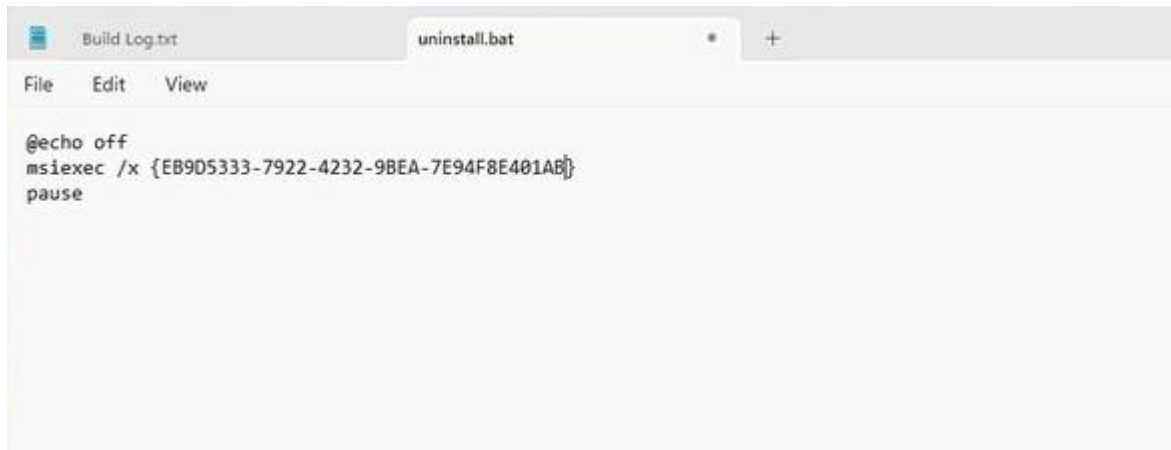
Create a batch file named:

uninstall.bat

Add the following script:

```
@echo off  
msiexec /x {ProductCode}  
pause
```

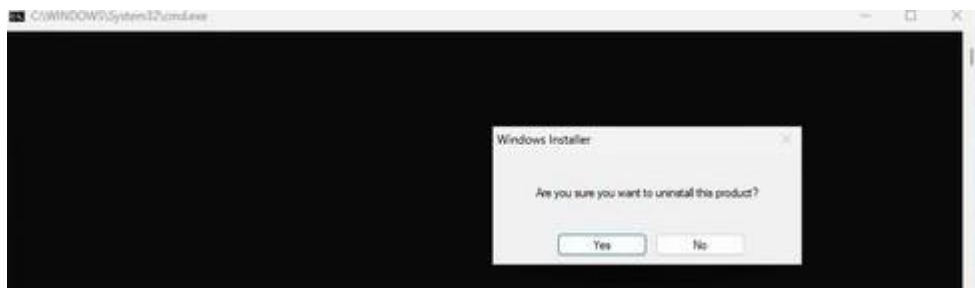
Image 2: Batch script



Step 4: Run the Script

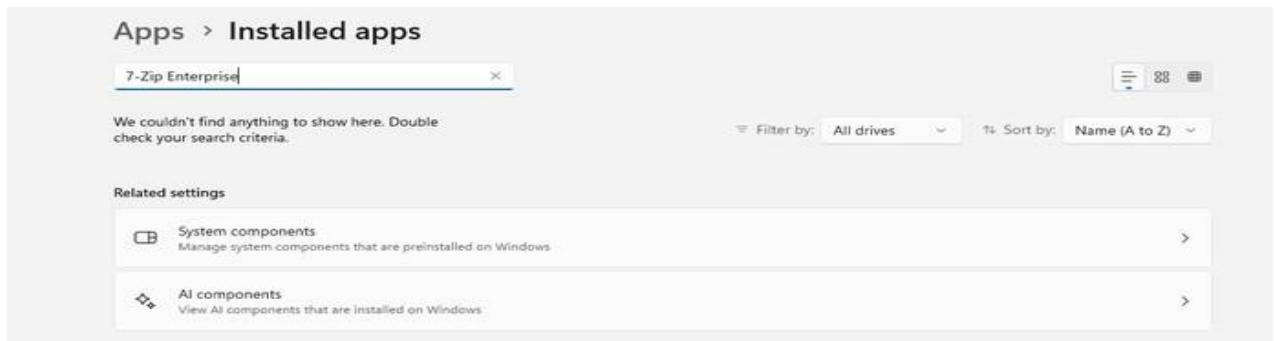
- Right-click **uninstall.bat**
- Select **Run as Administrator**

Image 3: Uninstall process started



Step 5: Verify Removal

Check installed applications.



The **7-Zip Enterprise** program should no longer appear in the list.

12.5 Result

The uninstall script successfully removed the packaged application using the Windows Installer command-line utility. The application and its related components were removed from the system.
